HWRGEFBTPALUG

Hardware Rework Guide for EdgeFast Bluetooth Protocol Abstraction Layer

Rev. 5 — 28 November 2022 **User manual**

Document information

Information	Content
Keywords	Hardware Rework, EdgeFast, Bluetooth, PAL, Protocol, Abstraction, Layer
Abstract	This document is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on various boards.

1 Hardware Rework Guide for MIMXRT1060-EVKB and AW-AM510MA

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVKB board and AW-AM510MA. The AW-AM510MA user guide is available here.

The hardware rework has three parts:

- HCI UART rework
- I2S2 rework
- M.2 SDIO rework

1.1 Hardware rework

- **HCI UART rework**
  1. Remove R293 and R354, connect R293 pin2 with R354 Pin1.
  2. Remove R241 and R163, connect R93 pin1 with R241 Pin2.
  3. Solder R96, R93, R87, R79, R70, and R345.
  4. Remove R193.

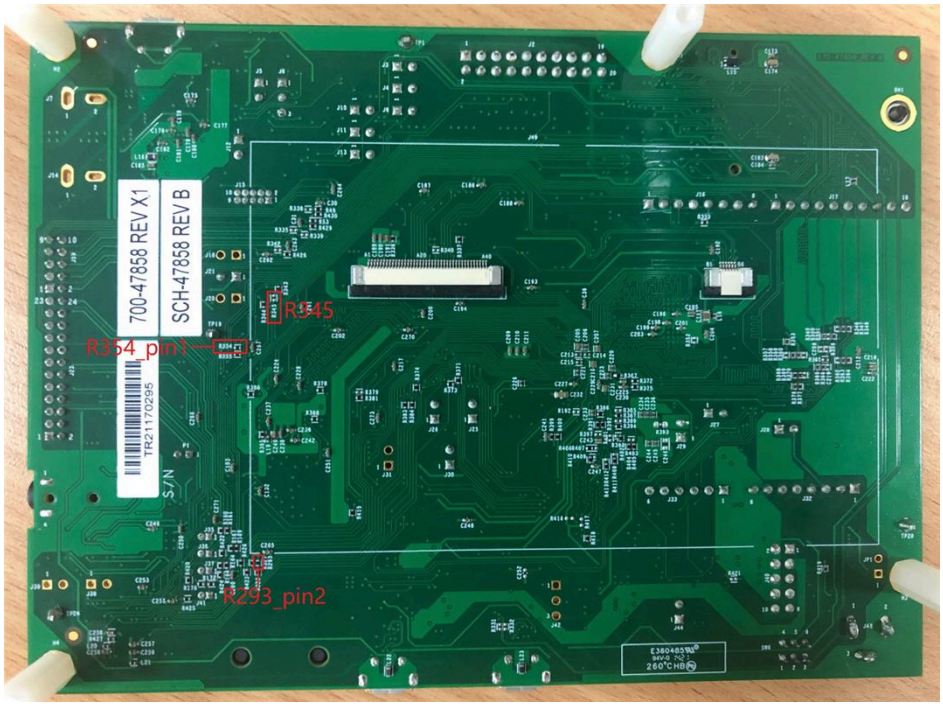


Figure 1. MIMXRT1060-EVKB (Back)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

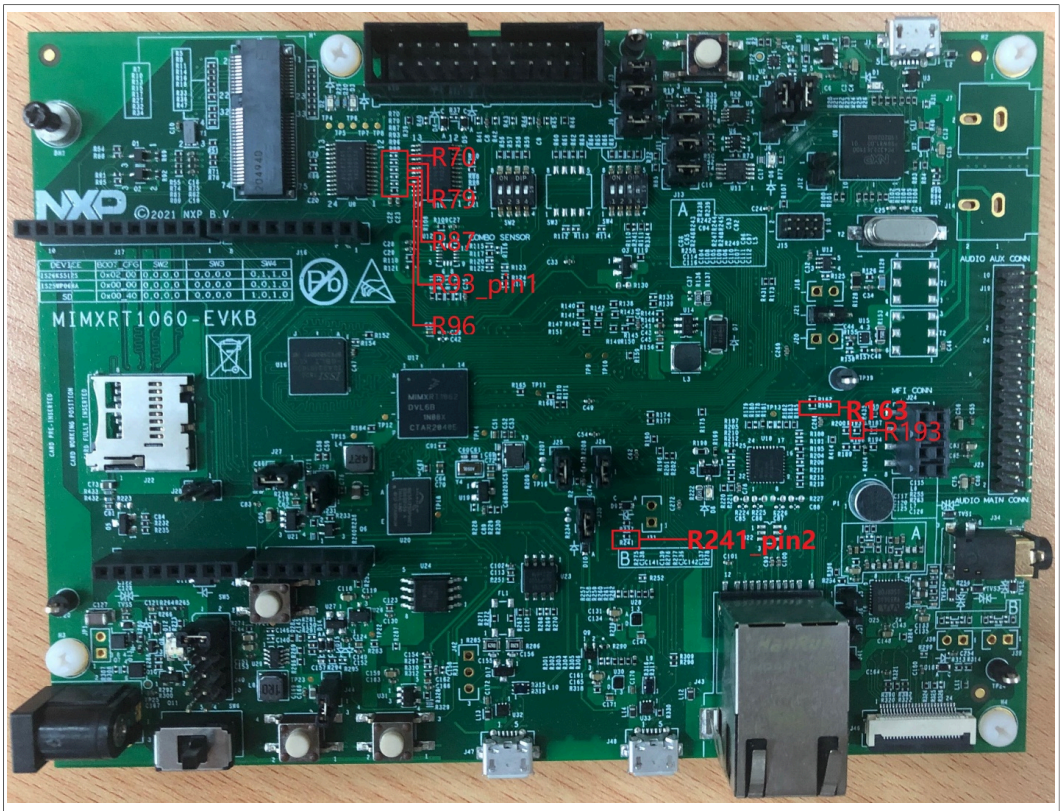


Figure 2. MIMXRT1060-EVKB (Front)

- **I2S2 rework**
  1. Open jumpers: J35, J36, J37, and J41.
  2. Connect J35 (Pin2) with J19 (Pin3) and solder R341.
  3. Connect J36 (Pin2) with J19 (Pin9) and solder R334
  4. Connect J41 (Pin2) with TP11.
  5. Connect J37 (Pin2) with J16 (Pin5).
  6. Connect R428 with R254 (Pin2), remove R254, R173, and R175.
  7. Remove R86, R76, and R381.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

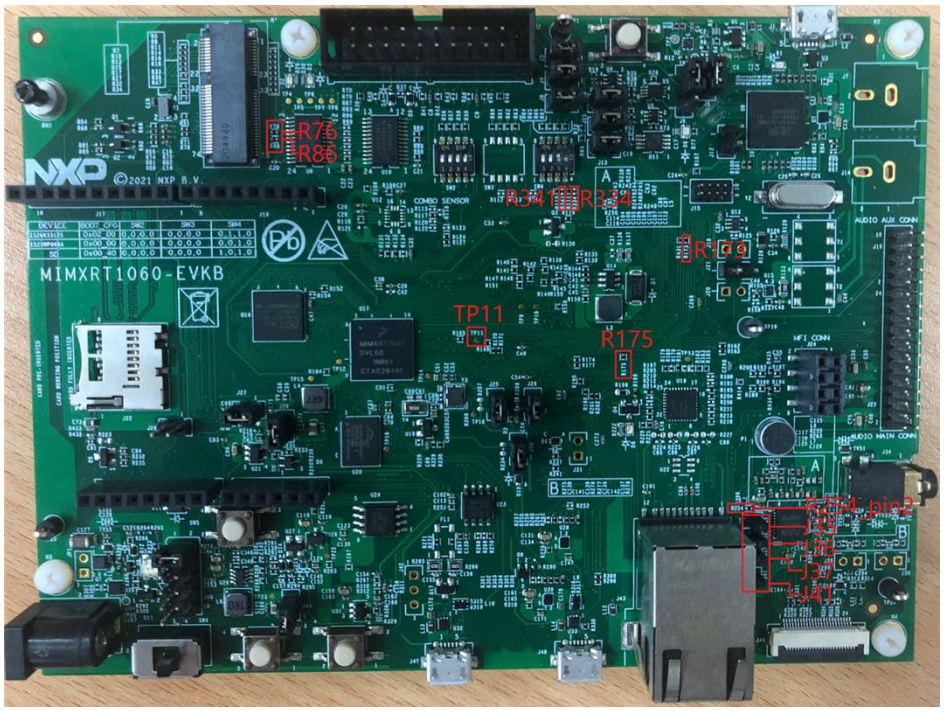


Figure 3. MIMXRT1060-EVKB (Front)

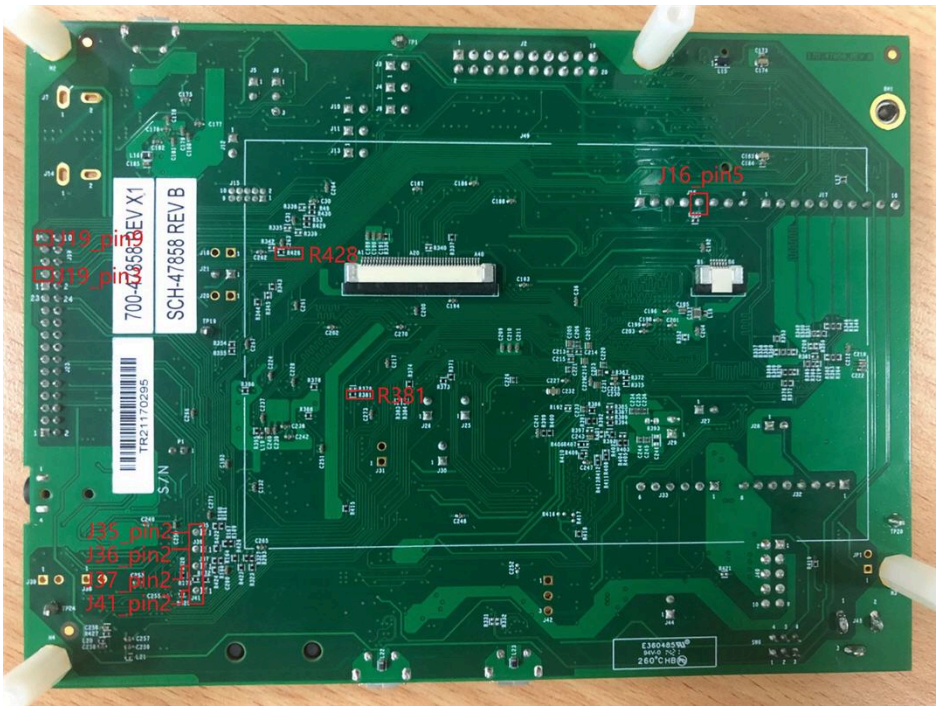


Figure 4. MIMXRT1060-EVKB (Back)

- **M.2 SDIO rework**
  1. Solder R368, R376, R347, R349, R365, and R363.
  2. Remove R364, R366, R351, R348, R377, and R369.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

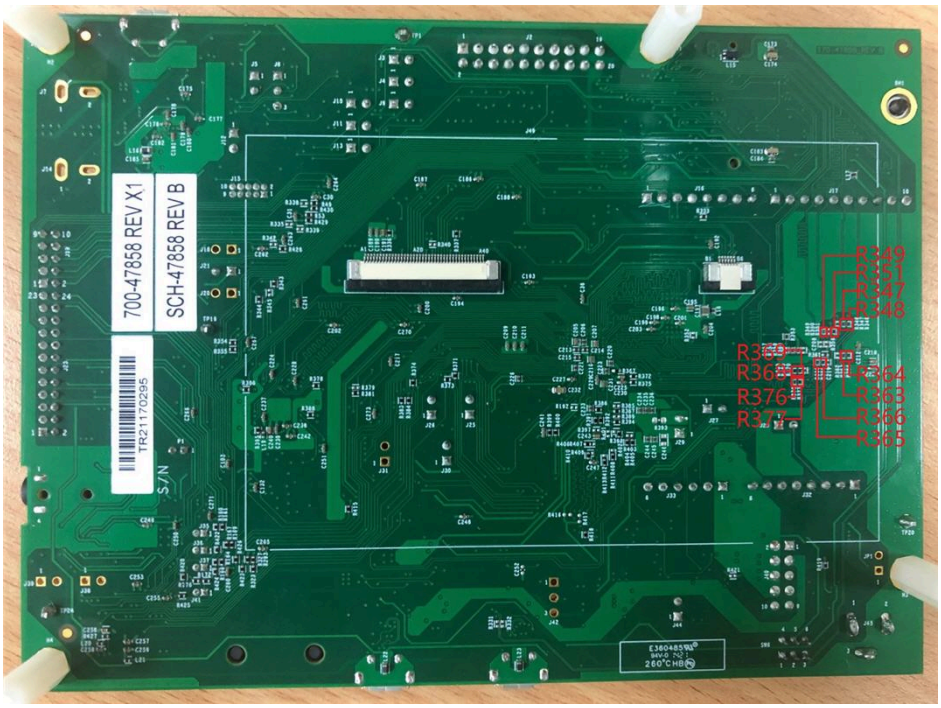


Figure 5. MIMXRT1060-EVKB (Back)

2 Hardware Rework Guide for MIMXRT1060-EVKB and AW-CM358MA

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVKB board and AW-CM358MA. The AW-CM358MA user guide is available here.

The hardware rework has three parts:

- HCI UART rework
- I2S2 rework
- M.2 SDIO rework

2.1 Hardware rework

- **HCI UART rework**
  1. Remove R293 and R354, connect R293 pin2 with R354 Pin1.
  2. Remove R241 and R163, connect R93 pin1 with R241 Pin2.
  3. Solder R96, R93, R87, R79, R70, and R345.
  4. Remove R193.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

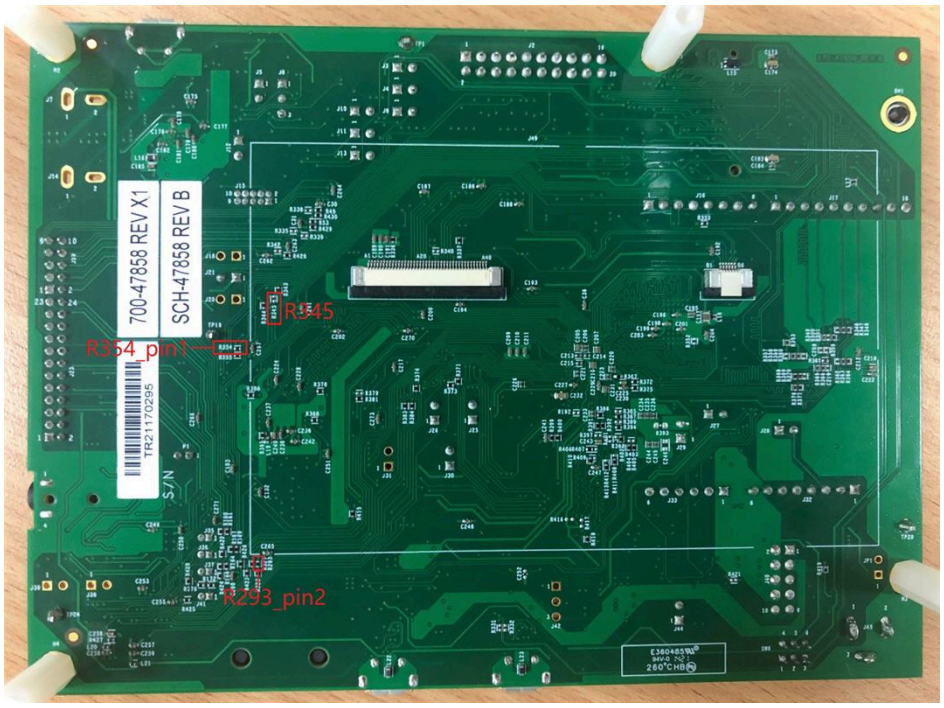


Figure 6. MIMXRT1060-EVKB (Back)

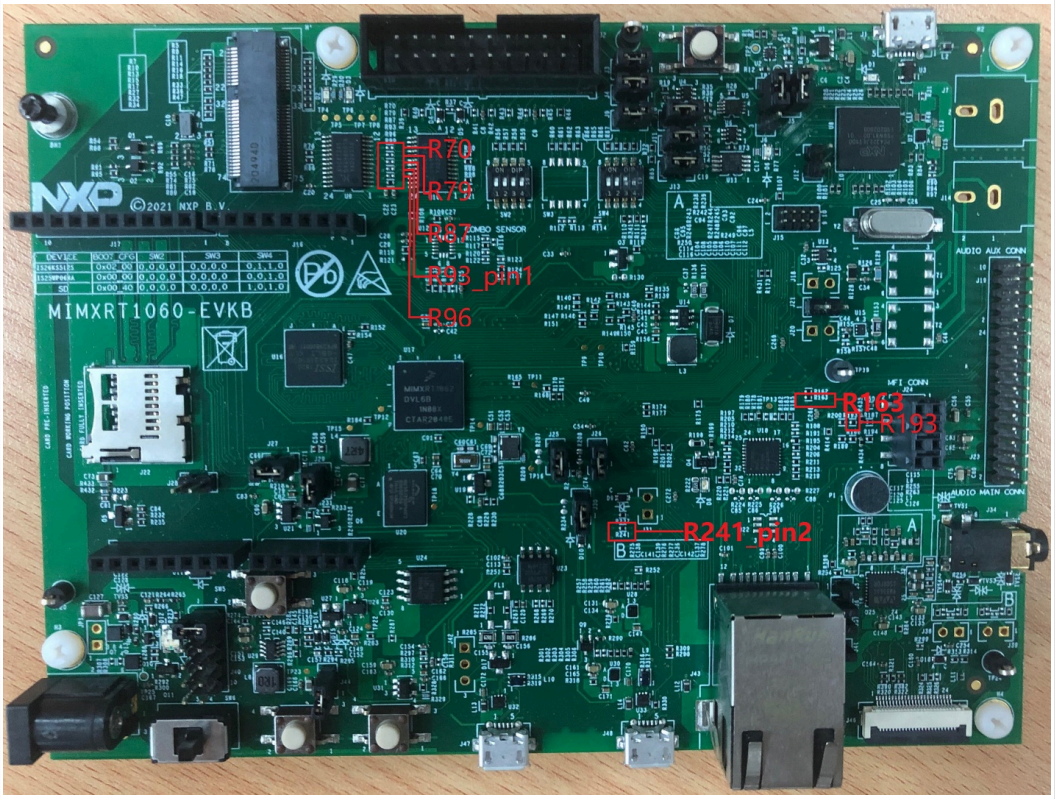


Figure 7. MIMXRT1060-EVKB (Front)

- **I2S2 rework**
  1. Open jumpers: J35, J36, J37, and J41.
  2. Connect J35 (Pin2) with J19 (Pin3) and solder R341.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

3. Connect J36 (Pin2) with J19 (Pin9) and solder R334.
4. Connect J41 (Pin2) with TP11.
5. Connect J37 (Pin2) with J16 (Pin5).
6. Connect R428 with R254 (Pin2), remove R254, R173, and R175.
7. Remove R86, R76, and R381.

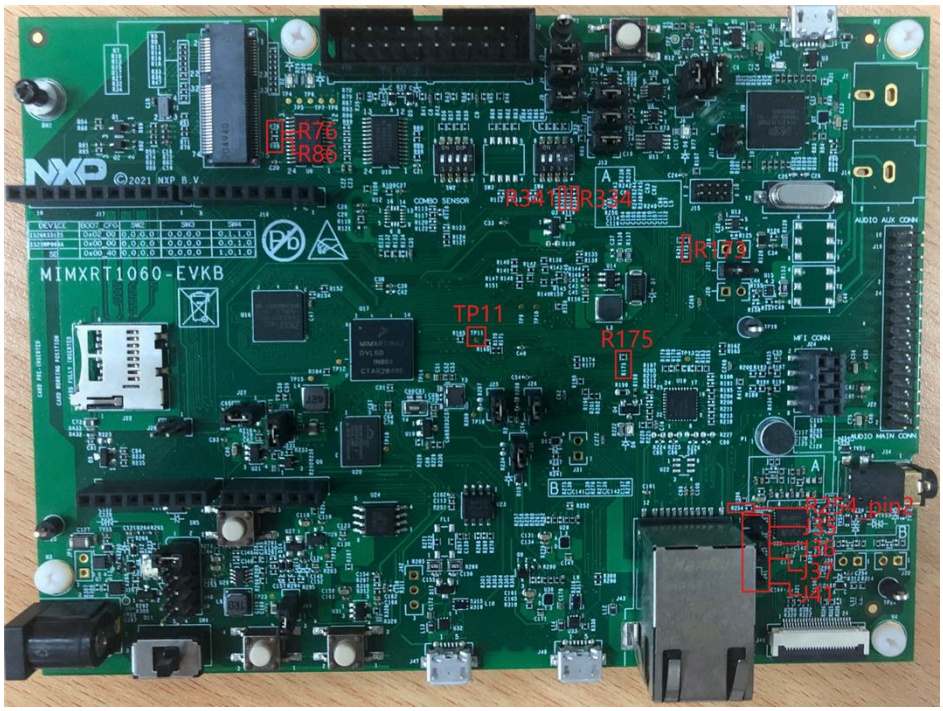


Figure 8. MIMXRT1060-EVKB (Front)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

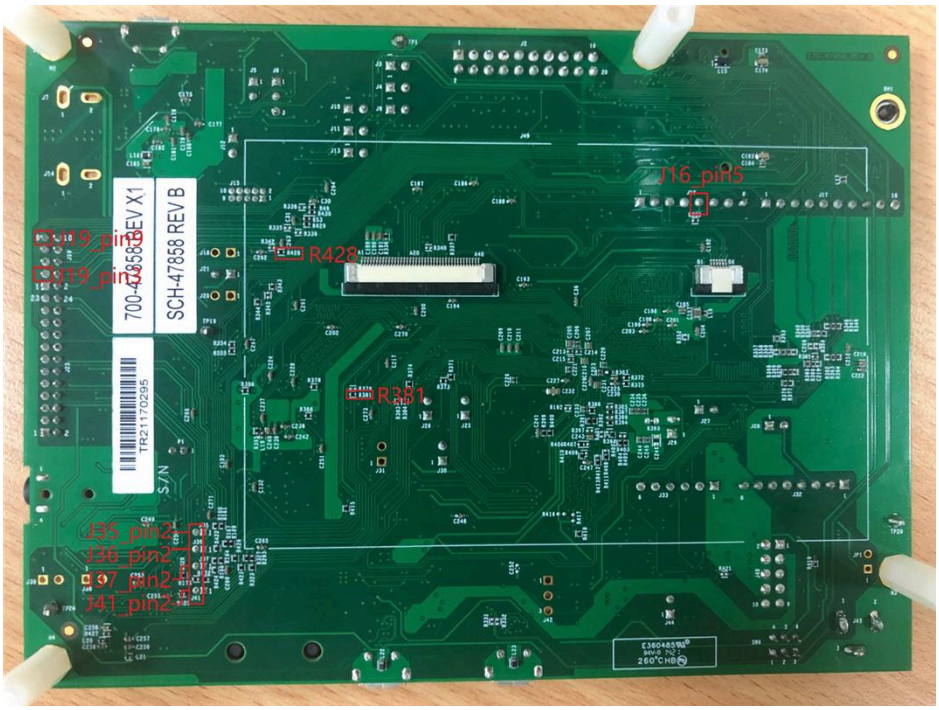


Figure 9. MIMXRT1060-EVKB (Back)

- **M.2 SDIO rework**
  1. Solder R368, R376, R347, R349, R365, and R363.
  2. Remove R364, R366, R351, R348, R377, and R369.

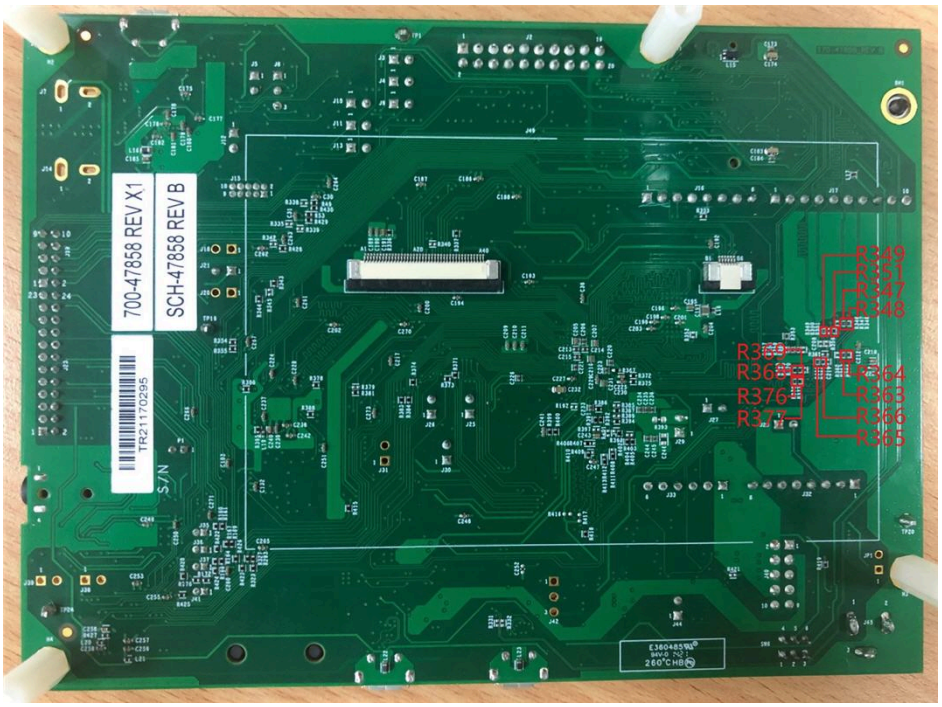


Figure 10. MIMXRT1060-EVKB (Back)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

3 Hardware Rework Guide for MIMXRT1060-EVKB and AW-AM457-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVKB board and AW-AM457-uSD. The AW-AM457-uSD user guide is available here.

The hardware rework has two parts:

- HCI UART port rework
- PCM interface rework

3.1 Hardware rework

- **HCI UART rework**
  Make sure resistors R368/R376/R347/R349/R365/R363/R193/R186 are removed. Connect the pins of two boards as the following table.

Table 1. Connect pins

Pin Name	AW-AM457-uSD	i.MXRT1060	PIN NAME	GPIO NAME of RT1060
UART_TXD	J10 (pin 4)	J16 (pin 1)	LPUART3_RXD	GPIO_AD_B1_07
UART_RXD	J10 (pin 2)	J16 (pin 2)	LPUART3_TXD	GPIO_AD_B1_06
UART_RTS	J10 (pin 6)	J33 (pin 3)	LPUART3_CTS	GPIO_AD_B1_04
UART_CTS	J10 (pin 8)	J33 (pin 4)	LPUART3_RTS	GPIO_AD_B1_05
GND	J6 (pin 7)	J32 (pin 7)	GND	GND

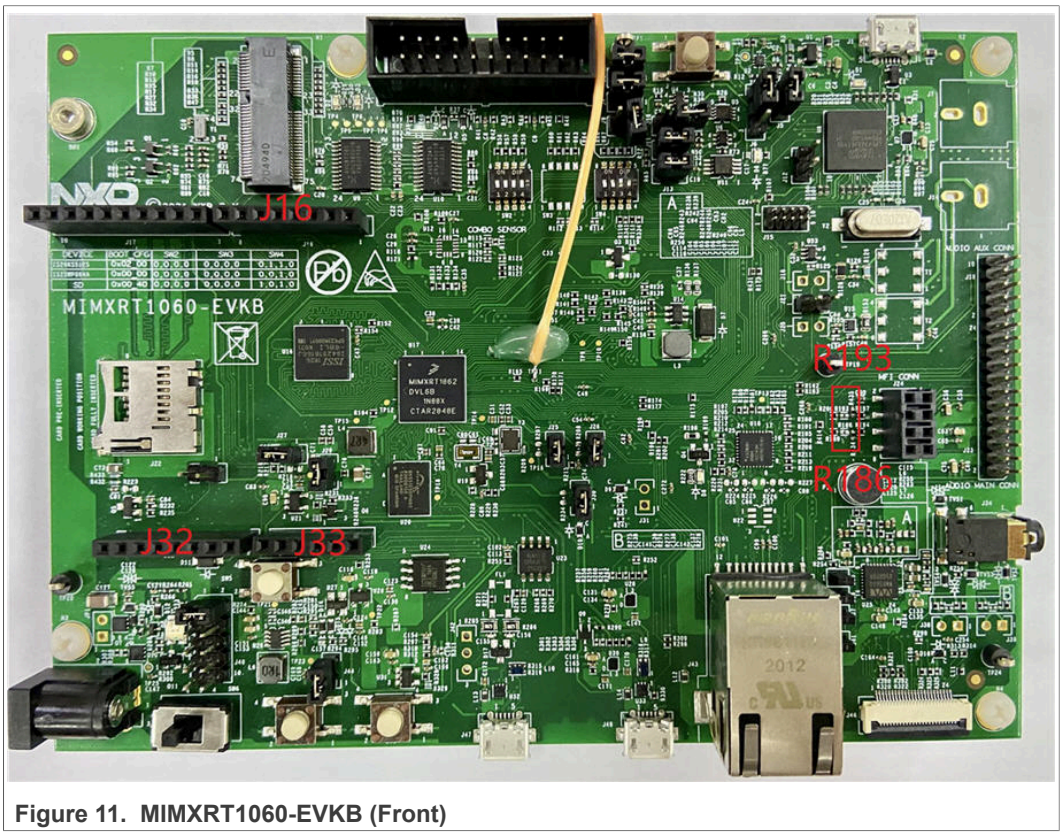


Figure 11. MIMXRT1060-EVKB (Front)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

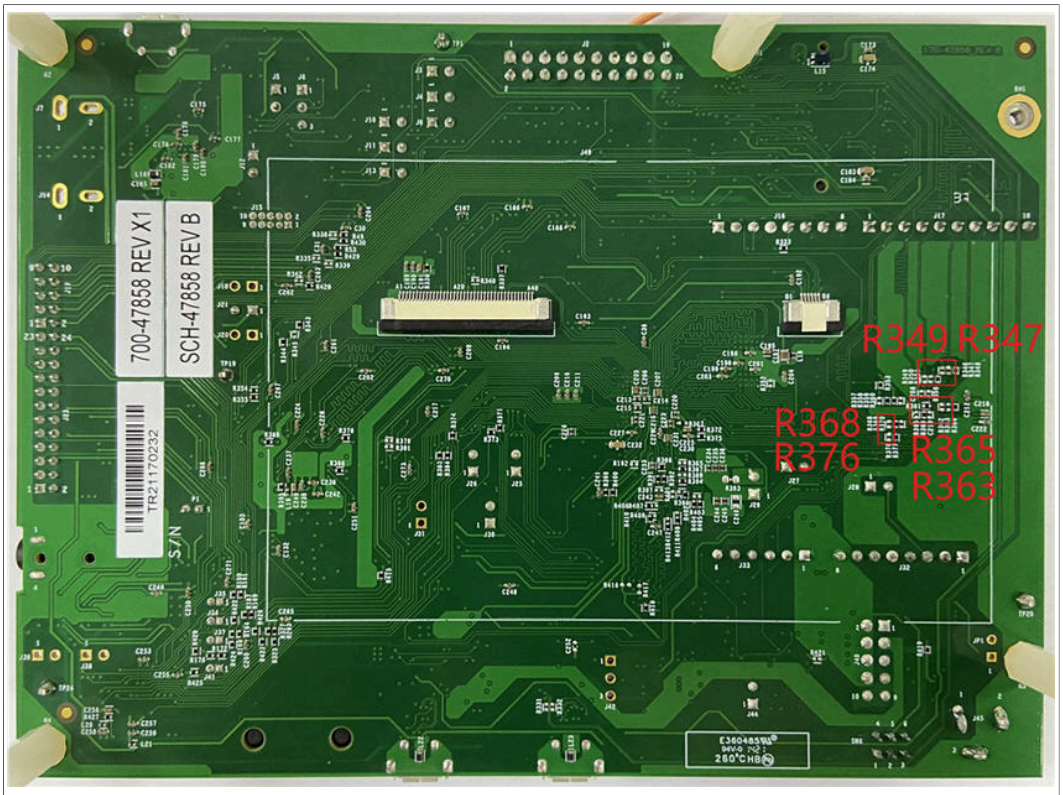


Figure 12. MIMXRT1060-EVKB (Back)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

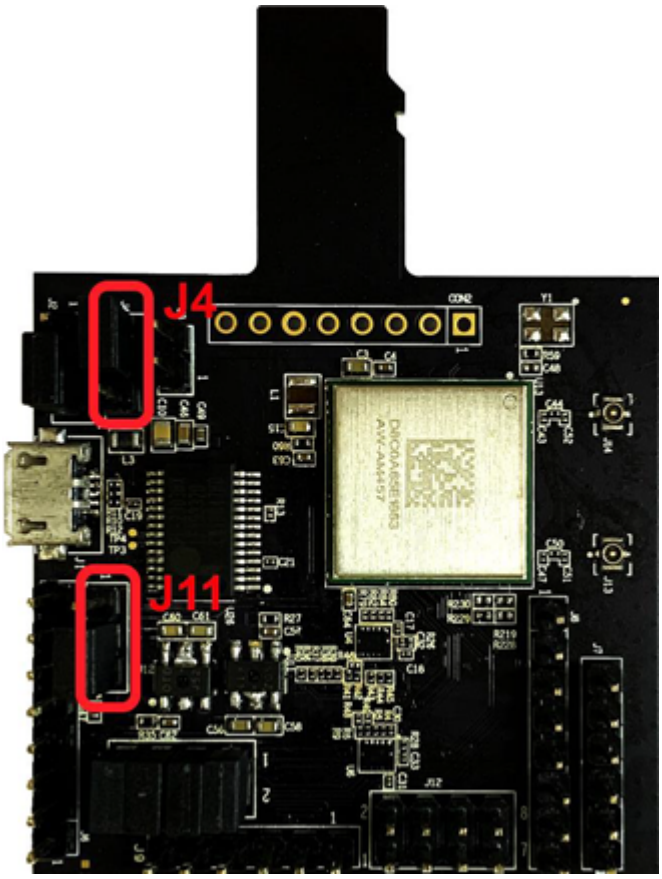


Figure 13. AW-AM457-uSD

Jumper Setting:

– Connect J4[2-3] for VIO 3.3 V supply

– Connect J11[2-3] for VIO_SD 3.3 V supply

- **PCM interface rework**

  Connect the pins of two boards as the following table.

Table 2. Connect pins

PIN NAME	AW-AM457-USD	I.MXRT1060	PIN NAME OF RT1060	GPIO NAME of RT1060
PCM_IN	J9 (pin 1)	J16 (pin 5)	SAI2_TXD	GPIO_AD_B0_09
PCM_OUT	J9 (pin 2)	TP11	SAI2_RXD	GPIO_AD_B0_08
PCM_SYNC	J9 (pin 3)	J2 (pin 9)	SAI2_RX_SYNC	GPIO_AD_B0_07
PCM_CLK	J9 (pin 4)	J10 (pin 2)	SAI2_RX_BCLK	GPIO_AD_B0_06
GND	J9 (pin 6)	J2 (pin 20)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

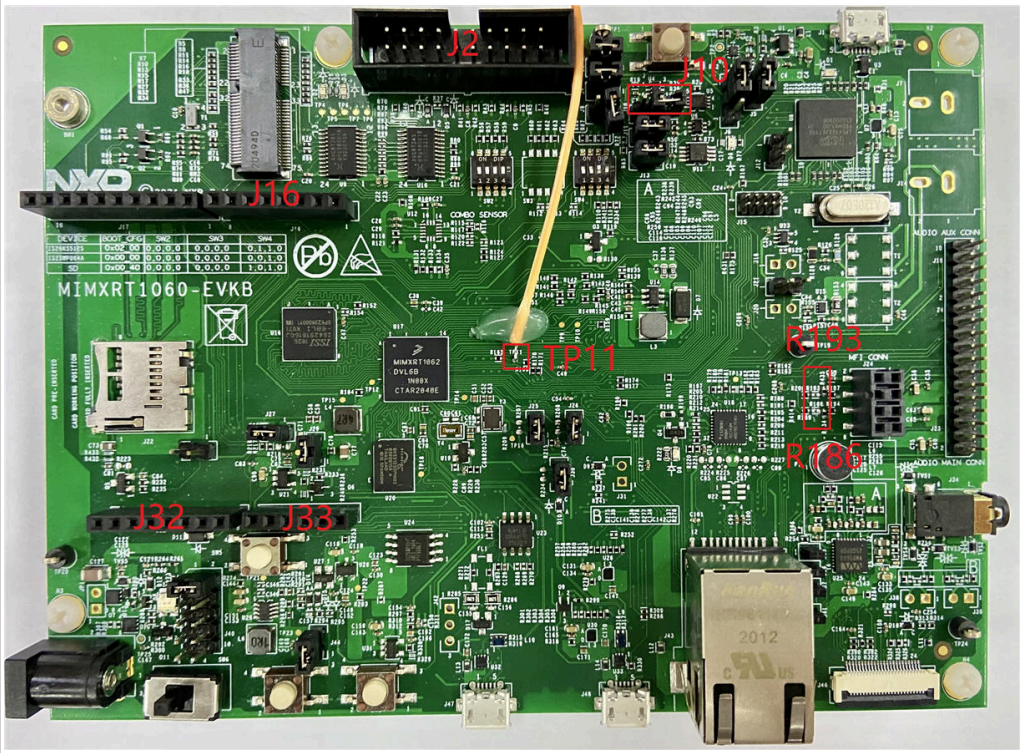


Figure 14. MIMXRT1060-EVKB (Front)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

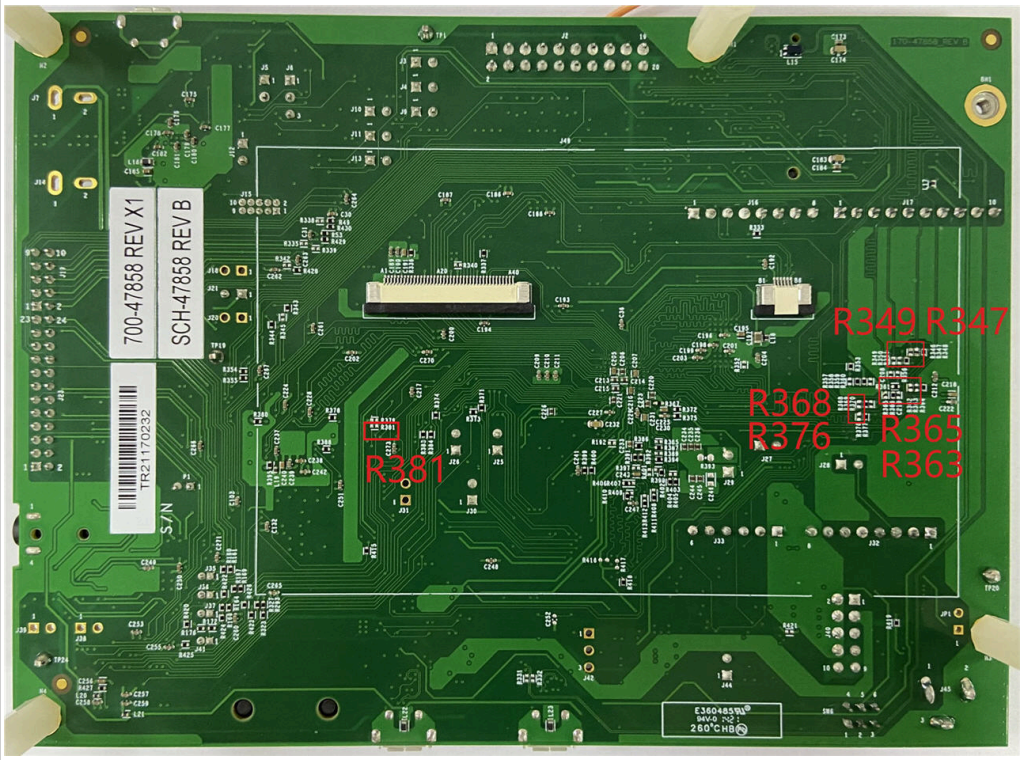


Figure 15. MIMXRT1060-EVKB (Back)

Note:
To support HFP feature, you must remove R381 on MIMXRT1060-EVKB.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

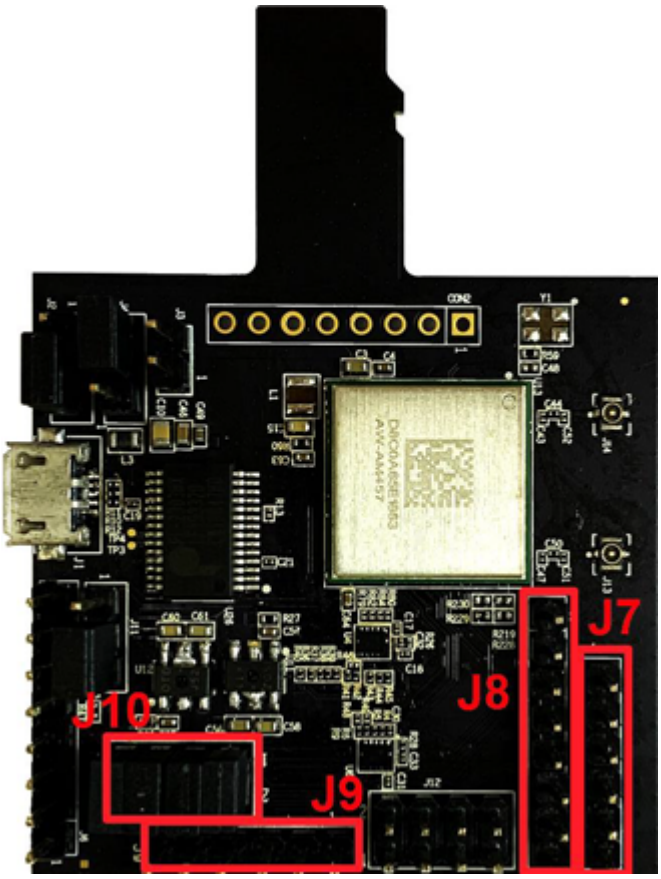


Figure 16. AW-AM457-uSD

4 Hardware Rework Guide for MIMXRT1060-EVKB and AW-CM358-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVKB board and AW-CM358-uSD. The AW-CM358-uSD user guide is available here.

The hardware rework has two parts:

- HCI UART port rework
- PCM interface rework

4.1 Hardware rework

- **HCI UART rework**
  Connect the pins of two boards as the following table.
  Make sure that the resistors R368/R376/R347/R349/R365/R363/R193/R186 are removed.

Table 3. Connect pins

Pin Name	AW-CM358-USD	i.MXRT1060	PIN NAME	GPIO NAME of RT1060
UART_TXD	J10 (pin 4)	J16 (pin 1)	LPUART3_RXD	GPIO_AD_B1_07

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 3. Connect pins*...continued*

Pin Name	AW-CM358-USD	i.MXRT1060	PIN NAME	GPIO NAME of RT1060
UART_RXD	J10 (pin 2)	J16 (pin 2)	LPUART3_TXD	GPIO_AD_B1_06
UART_RTS	J10 (pin 6)	J33 (pin 3)	LPUART3_CTS	GPIO_AD_B1_04
UART_CTS	J10 (pin 8)	J33 (pin 4)	LPUART3_RTS	GPIO_AD_B1_05
GND	J6 (pin 7)	J32 (pin 7)	GND	GND

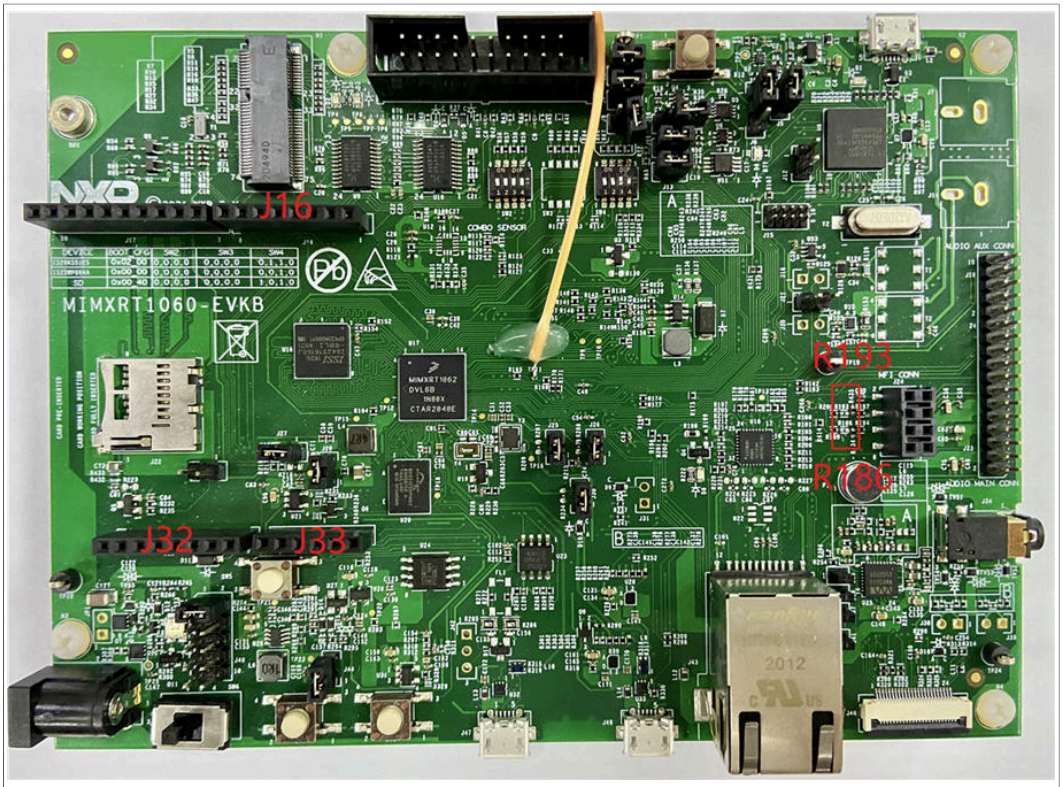


Figure 17. MIMXRT1060-EVKB (Front)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

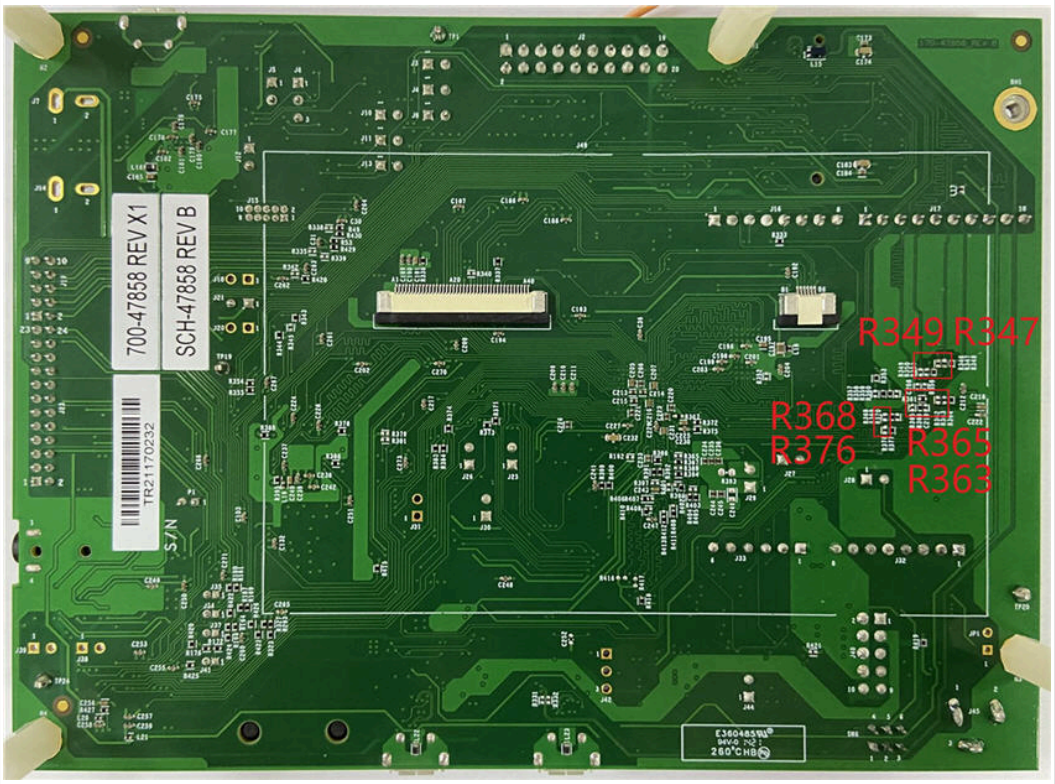


Figure 18. MIMXRT1060-EVKB (Back)

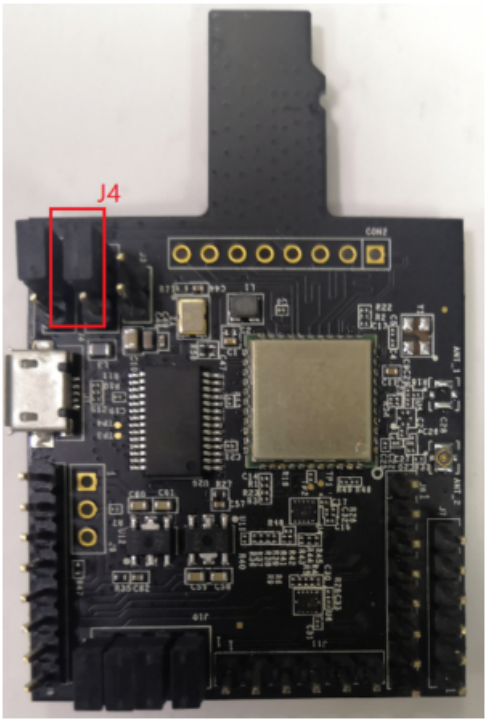


Figure 19. AW-CM358-uSD

Jumper Setting:
Connect J4[1-2] for VIO 1.8 V supply.

- **PCM interface rework**

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Connect the pins of two boards as the following table.

Table 4. Connect pins

PIN NAME	AW-CM358-USD	I.MXRT1060	PIN NAME OF RT1060	GPIO NAME of RT1060
PCM_IN	J11 (pin 1)	J16 (pin 5)	SAI2_TXD	GPIO_AD_B0_09
PCM_OUT	J11 (pin 2)	TP11	SAI2_RXD	GPIO_AD_B0_08
PCM_SYNC	J11 (pin 3)	J2 (pin 9)	SAI2_RX_SYNC	GPIO_AD_B0_07
PCM_CLK	J11 (pin 4)	J10 (pin 2)	SAI2_RX_BCLK	GPIO_AD_B0_06
GND	J11 (pin 6)	J2 (pin 20)	GND	GND

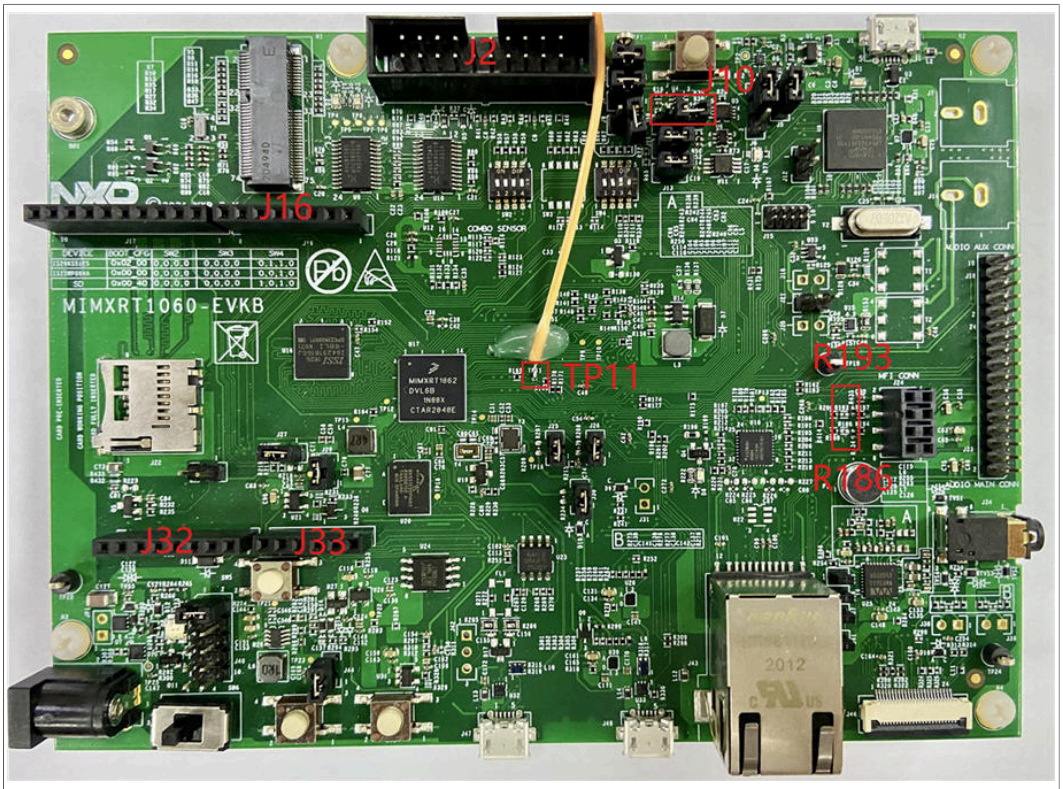


Figure 20. MIMXRT1060-EVKB (Front)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

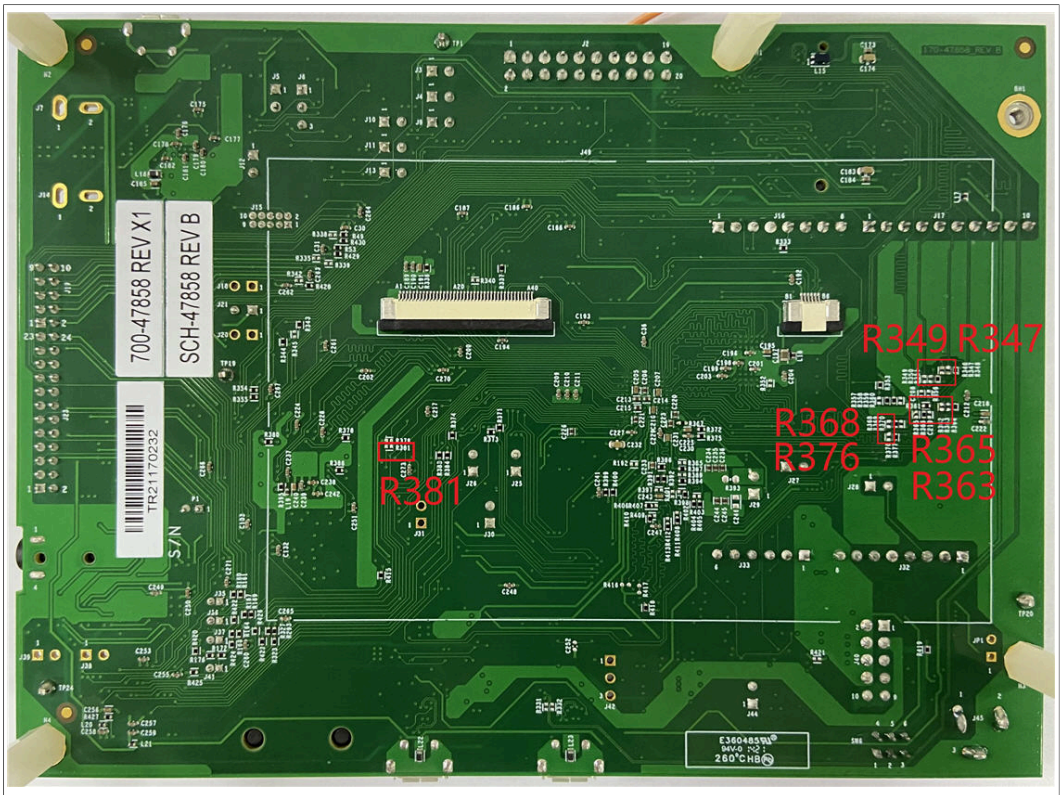


Figure 21. MIMXRT1060-EVKB (Back)

Note:

To support HFP feature, you must remove R381 on MIMXRT1060-EVKB.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

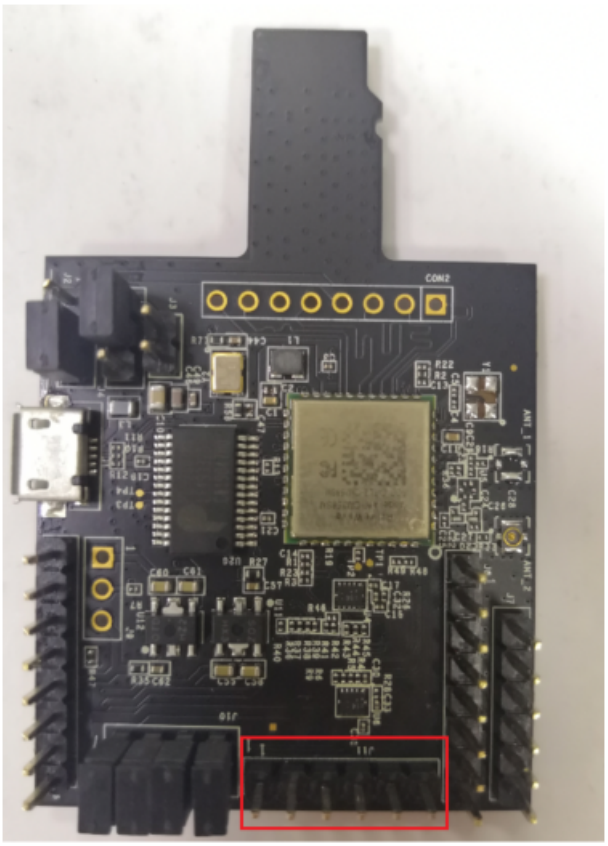


Figure 22. AW-CM358-uSD

5 Hardware Rework Guide for MIMXRT1060-EVKB and Murata uSD-M.2 Adapter

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVKB board and Murata uSD-M.2 adapter. For details on the Murata uSD-M.2 Adapter, see the Murata uSD-M.2 webpage here. The hardware rework consists of three parts:

- Murata uSDM
- 2 jumper settings
- HCI UART and PCM interface rework

5.1 Hardware rework

- **Murata uSD-M.2 jumper settings**
  - J12 = 1-2: WLAN-SDIO & BT-PCM = 1.8 V
  - J13 = 1-2: BT-UART & WLAN/BT-CTRL = 3.3 V
  - J1 = 2-3: 3.3 V from uSD connector
- **HCI UART interface rework**
  Connect the TX/RX/RTS/CTS pins of the two boards as show in Table 5using the jumper cables included in the Murata's uSD-M.2 Adapter kit.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 5. Connect the TX/RX/RTS/CTS pins

Pin name	uSD-M.2 adapter pin	i.MX RT1060-EVKB pin	Pin name of RT1060-EVKB	GPIO name of RT1060-EVKB
BT_UART_TXD_ HOST	J9 (pin 1)	J16 (pin 1)	LPUART3_RXD	GPIO_AD_B1_07
BT_UART_RXD_ HOST	J9 (pin 2)	J16 (pin 2)	LPUART3_TXD	GPIO_AD_B1_06
BT_UART_RTS_ HOST	J8 (pin 3)	J33 (pin 3)	LPUART3_CTS	GPIO_AD_B1_04
BT_UART_CTS_ HOST	J8 (pin 4)	J33 (pin 4)	LPUART3_RTS	GPIO_AD_B1_05
GND	J7 (pin 7)	J32 (pin 7)	GND	GND

- **PCM interface rework**
  Connect the PCM_IN/PCM_OUT/PCM_SYNC/PCM_CLK/GND pins of the two boards as shown in Table 6.

Table 6. Connect the PCM_IN/PCM_OUT/PCM_SYNC/PCM_CLK/GND pins

Pin name	uSD-M.2 adapter pin	i.MXRT1060-EVKB pin	Pin name of RT1060-EVKB	GPIO name of RT1060-EVKB
PCM_IN	J5 (pin1)	J16 (pin5)	SAI2_TXD	GPIO_AD_B0_09
PCM_OUT	J5 (pin3)	TP11	SAI2_RXD	GPIO_AD_B0_08
PCM_SYNC	J5 (pin5)	J2 (pin9)	SAI2_RX_SYNC	GPIO_AD_B0_07
PCM_CLK	J5 (pin7)	J10 (pin2)	SAI2_RX_BCLK	GPIO_AD_B0_06
GND	J5 (pin15)	J2 (pin20)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

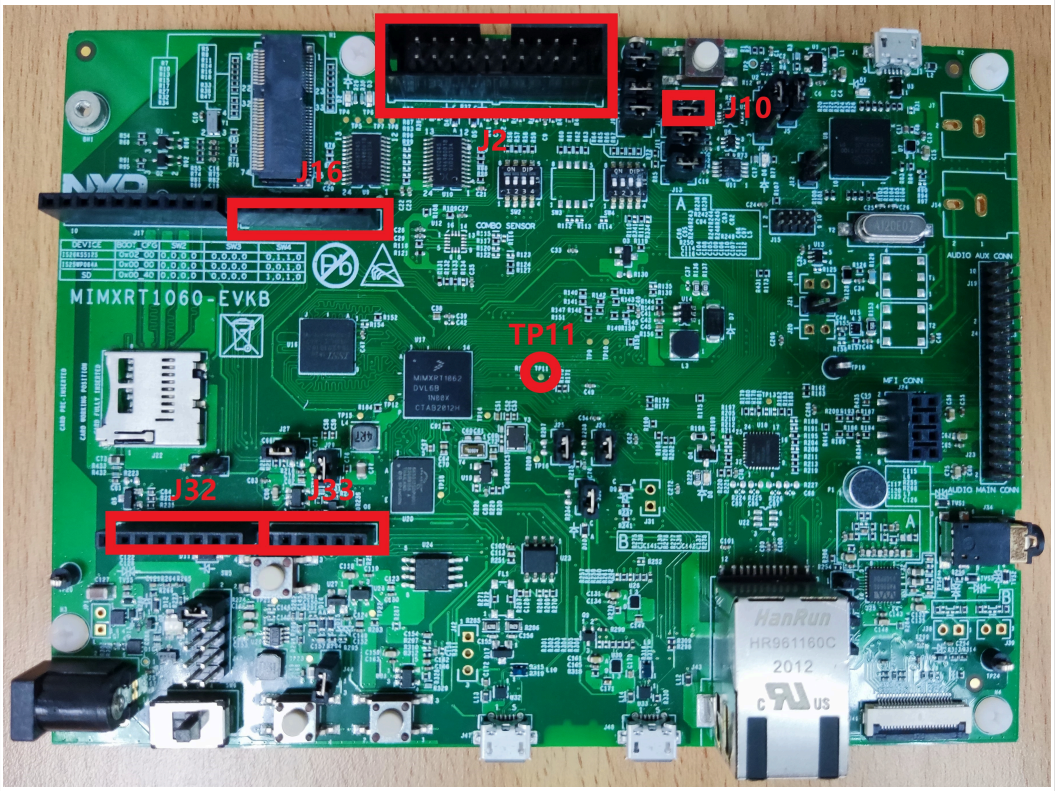


Figure 23. MIMXRT1060-EVKB board

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

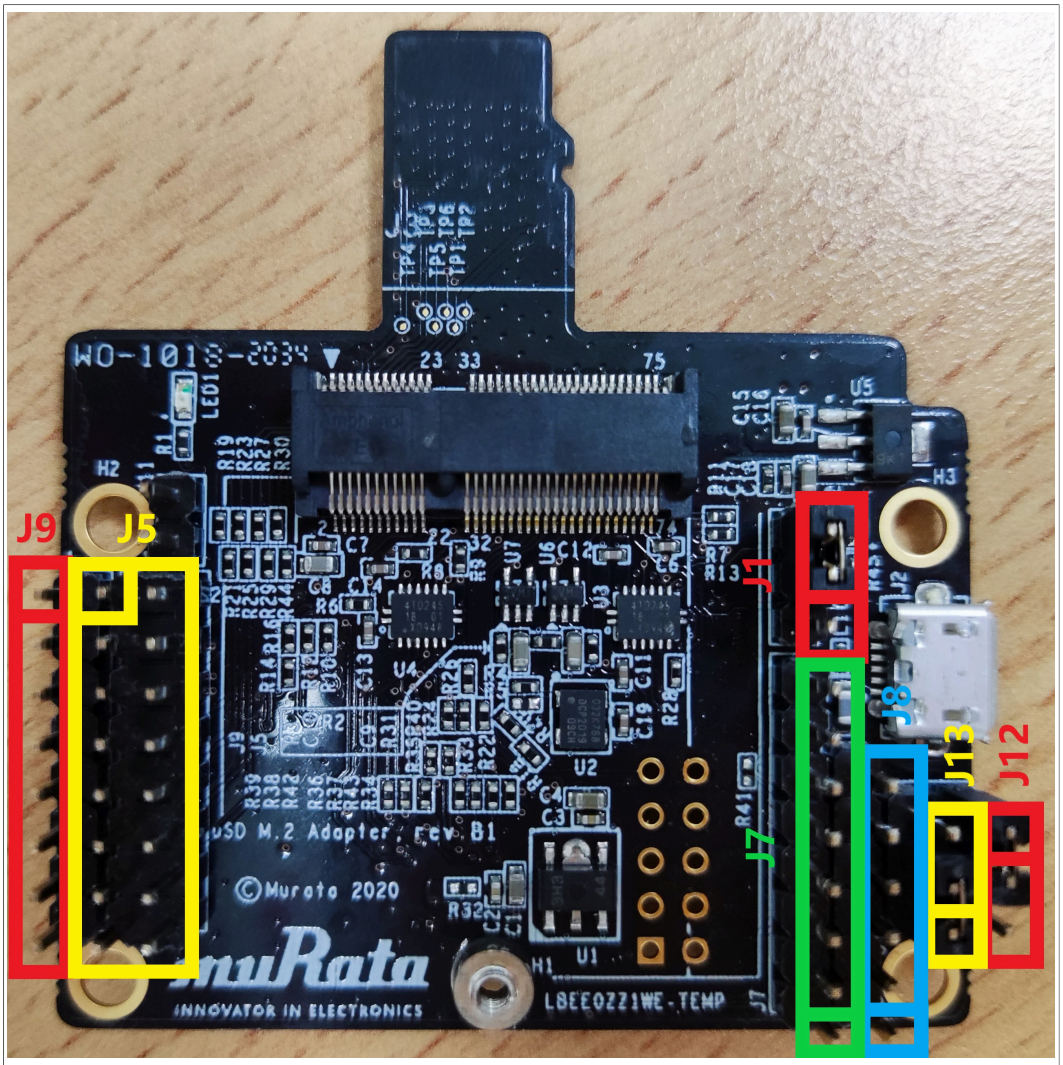


Figure 24. SD-M.2_Adapter

6 Hardware Rework Guide for MIMXRT1060-EVKB and AW-AM510-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVKB board and AW-AM510-uSD. The AW-AM510-uSD user guide is available here.

The hardware rework has two parts:

- HCI UART port rework
- PCM interface rework

6.1 Hardware rework

- **HCI UART rework**
  Make sure resistors R368/R376/R347/R349/R365/R363/R193/R186 are removed.
  Connect the pins of two boards as the following table.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 7. Connect pins

Pin Name	AW-AM510-uSD	i.MXRT1060	PIN NAME	GPIO NAME of RT1060
UART_TXD	J10 (pin 4)	J16 (pin 1)	LPUART3_RXD	GPIO_AD_B1_07
UART_RXD	J10 (pin 2)	J16 (pin 2)	LPUART3_TXD	GPIO_AD_B1_06
UART_RTS	J10 (pin 6)	J33 (pin 3)	LPUART3_CTS	GPIO_AD_B1_04
UART_CTS	J10 (pin 8)	J33 (pin 4)	LPUART3_RTS	GPIO_AD_B1_05
GND	J6 (pin 7)	J32 (pin 7)	GND	GND

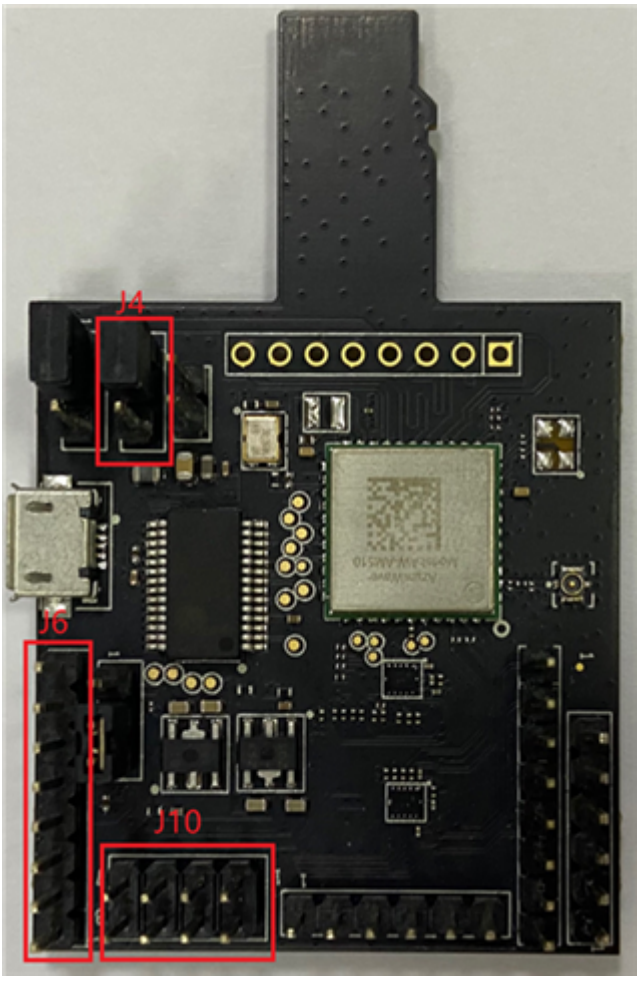


Figure 25. AW-AM510-uSD

Jumper Setting:

– Connect J4[2-3] for VIO 3.3 V supply

- **PCM interface rework**

  Connect the pins of two boards as the following table.

Table 8. Connect pins

PIN NAME	AW-AM510-USD	I.MXRT1060	PIN NAME OF RT1060	GPIO NAME of RT1060
PCM_IN	J11 (pin 1)	J16 (pin 5)	SAI2_TXD	GPIO_AD_B0_09
PCM_OUT	J11 (pin 2)	TP11	SAI2_RXD	GPIO_AD_B0_08

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 8. Connect pins*...continued*

PIN NAME	AW-AM510-USD	I.MXRT1060	PIN NAME OF RT1060	GPIO NAME of RT1060
PCM_SYNC	J11 (pin 3)	J2 (pin 9)	SAI2_RX_SYNC	GPIO_AD_B0_07
PCM_CLK	J11 (pin 4)	J10 (pin 2)	SAI2_RX_BCLK	GPIO_AD_B0_06
GND	J11 (pin 6)	J2 (pin 20)	GND	GND

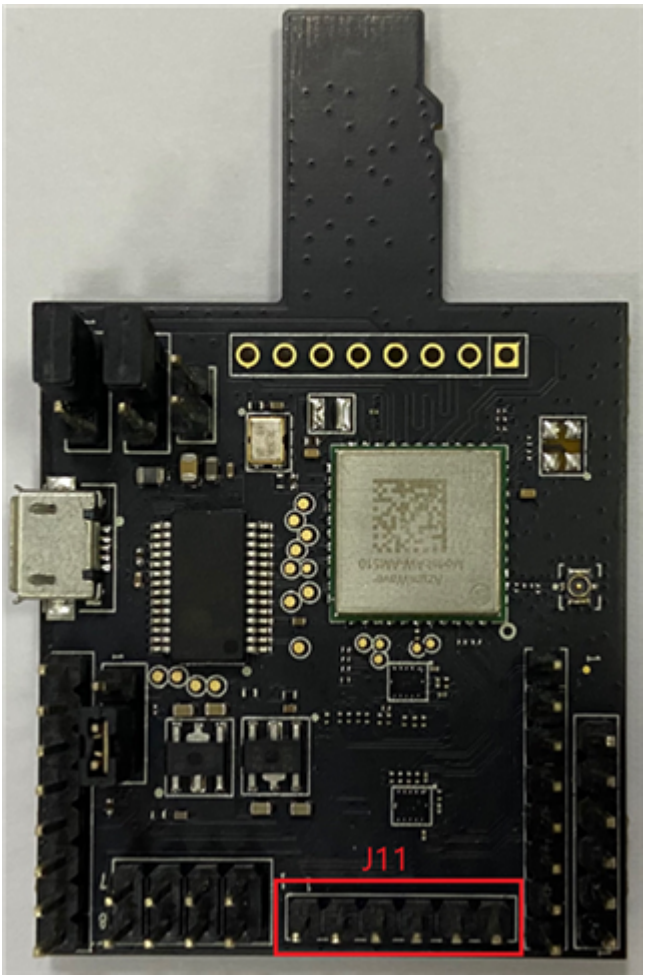


Figure 26. AW-AM510-uSD

7 Hardware Rework Guide for MIMXRT1060-EVK and AW-AM457-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVK board and AW-AM457-uSD. The AW-AM457-uSD user guide is available here.

The hardware rework has two parts:

- HCI UART port rework
- PCM interface rework

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

7.1 Hardware rework

- **HCI UART rework**
  Connect the pins of two boards as the following table.

Table 9. Connect pins

Pin Name	AW-AM457-uSD	i.MXRT1060	PIN NAME	GPIO NAME of RT1060
UART_TXD	J10 (pin 4)	J22 (pin 1)	LPUART3_RXD	GPIO_AD_B1_07
UART_RXD	J10 (pin 2)	J22 (pin 2)	LPUART3_TXD	GPIO_AD_B1_06
UART_RTS	J10 (pin 6)	J23 (pin 3)	LPUART3_CTS	GPIO_AD_B1_04
UART_CTS	J10 (pin 8)	J23 (pin 4)	LPUART3_RTS	GPIO_AD_B1_05
GND	J6 (pin 7)	J25 (pin 7)	GND	GND

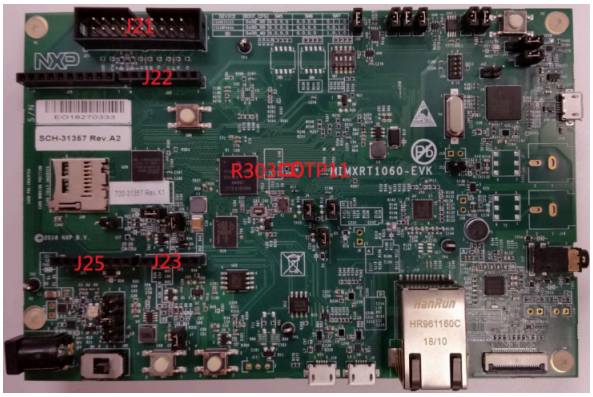


Figure 27. MIMXRT1060-EVK

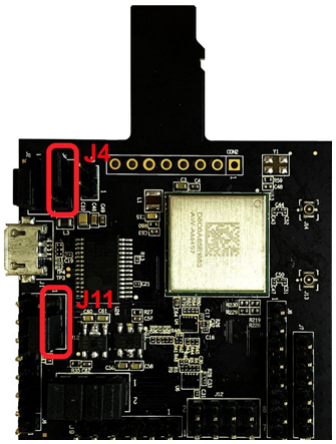


Figure 28. AW-AM457-uSD

Jumper Setting:

– Connect J4[2-3] for VIO 3.3 V supply

– Connect J11[2-3] for VIO_SD 3.3 V supply

- **PCM interface rework**
  Connect the pins of two boards as the following table.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 10. Connect pins

PIN NAME	AW-AM457-USD	I.MXRT1060	PIN NAME	GPIO NAME of RT1060
PCM_IN	J9 (pin 1)	J22 (pin 5)	SAI2_TXD	GPIO_AD_B0_09
PCM_OUT	J9 (pin 2)	TP11	SAI2_RXD	GPIO_AD_B0_08
PCM_SYNC	J9 (pin 3)	J21 (pin 9)	SAI2_RX_SYNC	GPIO_AD_B0_07
PCM_CLK	J9 (pin 4)	J21 (pin 7)	SAI2_RX_BCLK	GPIO_AD_B0_06
GND	J9 (pin 6)	J21 (pin 20)	GND	GND

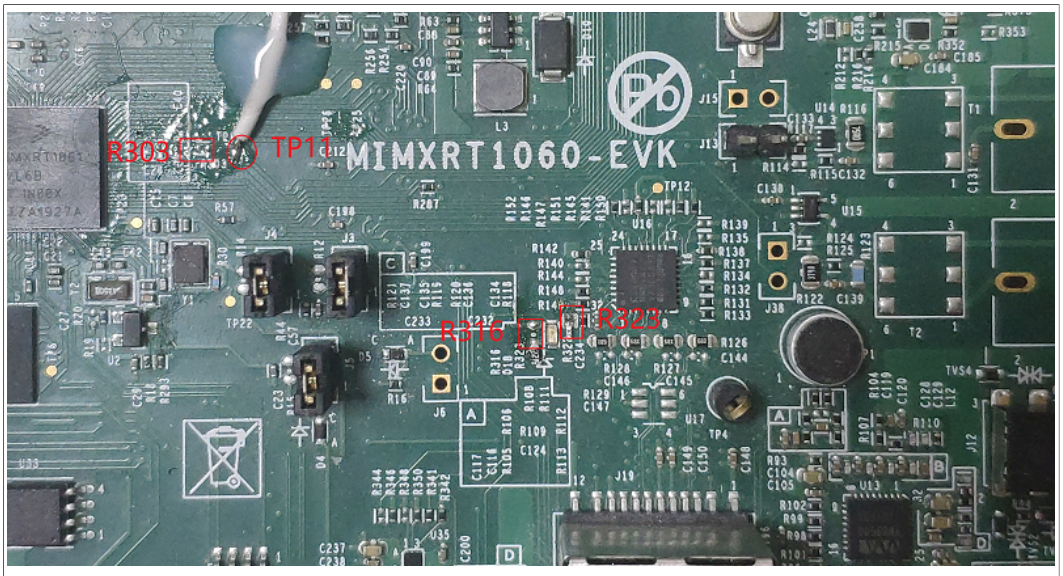


Figure 29. MIMXRT1060-EVK

Note:

To support HFP feature, you must remove R316 and R323 on MIMXRT1060-EVK.
To run HFP feature, you must remove J47 jumper.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

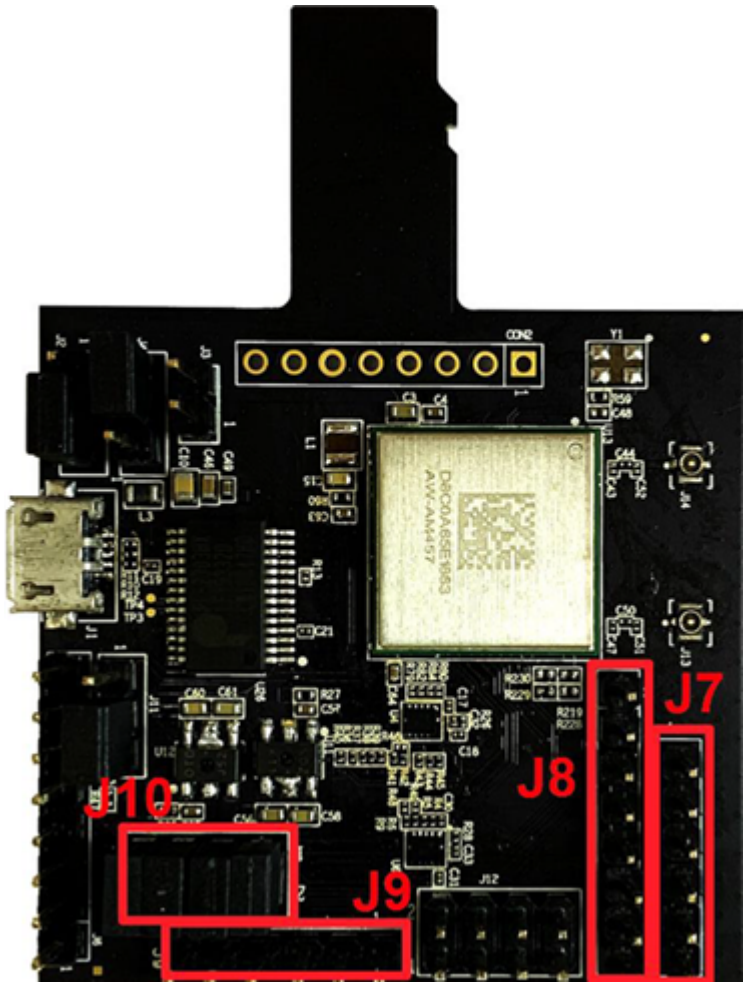


Figure 30. AW-AM457-uSD

8 Hardware Rework Guide for MIMXRT1060-EVK and AW-CM358-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVK board and AW-CM358-uSD. The AW-CM358-uSD user guide is available here.

The hardware rework has two parts:

- HCI UART port rework
- PCM interface rework

8.1 Hardware rework

1. **HCI UART rework**
   Connect the pins of two boards as the following table.

Table 11. Connect pins

Pin Name	AW-CM358-USD	i.MXRT1060	PIN NAME	GPIO NAME of RT1060
UART_TXD	J10 (pin 4)	J22 (pin 1)	LPUART3_RXD	GPIO_AD_B1_07

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 11. Connect pins*...continued*

Pin Name	AW-CM358-USD	i.MXRT1060	PIN NAME	GPIO NAME of RT1060
UART_RXD	J10 (pin 2)	J22 (pin 2)	LPUART3_TXD	GPIO_AD_B1_06
UART_CTS	J10 (pin 8)	J23 (pin 4)	LPUART3_RTS	GPIO_AD_B1_05
UART_RTS	J10 (pin 6)	J23 (pin 3)	LPUART3_CTS	GPIO_AD_B1_04
GND	J6 (pin 7)	J25 (pin 7)	GND	GND

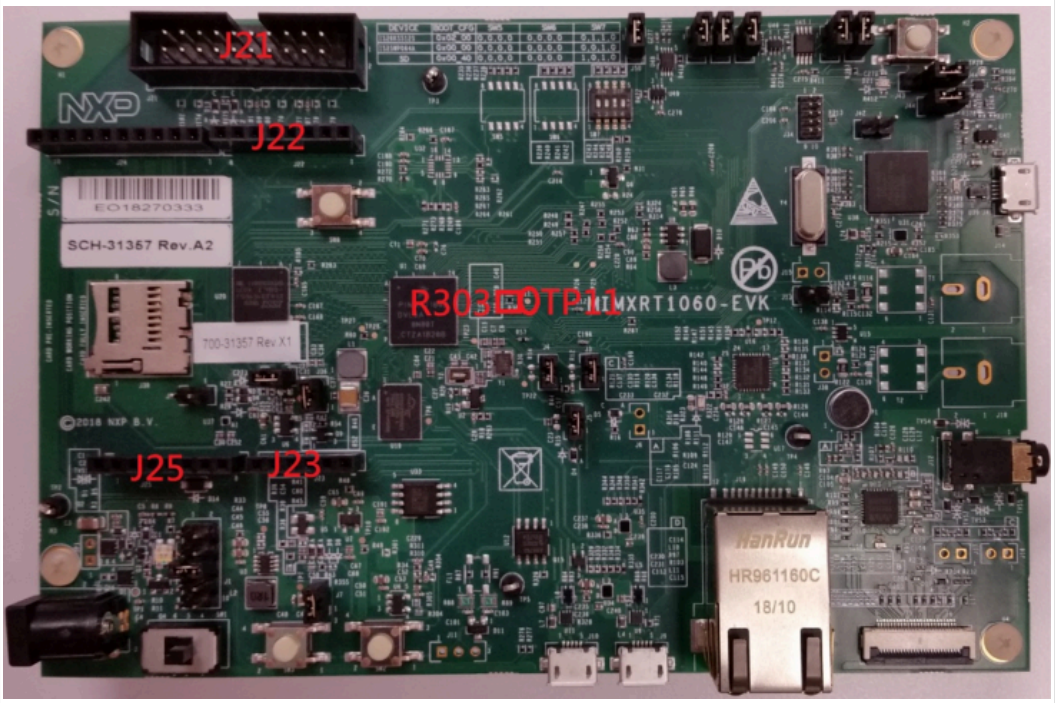


Figure 31. MIMXRT1060-EVK

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

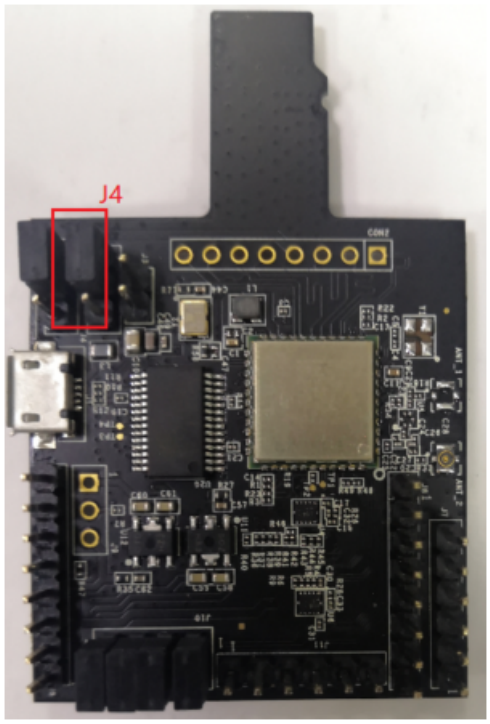


Figure 32. AW-CM358-uSD

Jumper Setting:
Connect J4[1-2] for VIO 1.8 V supply.

2. **PCM interface rework**
   Connect the pins of two boards as the following table.

Table 12. Connect pins

PIN NAME	AW-CM358-USD	I.MXRT1060	PIN NAME	GPIO NAME of RT1060
PCM_IN	J11 (pin 1)	J22 (pin 5)	SAI2_TXD	GPIO_AD_B0_09
PCM_OUT	J11 (pin 2)	TP11	SAI2_RXD	GPIO_AD_B0_08
PCM_SYNC	J11 (pin 3)	J21 (pin 9)	SAI2_RX_SYNC	GPIO_AD_B0_07
PCM_CLK	J11 (pin 4)	J21 (pin 7)	SAI2_RX_BCLK	GPIO_AD_B0_06
GND	J11 (pin 6)	J21 (pin 20)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

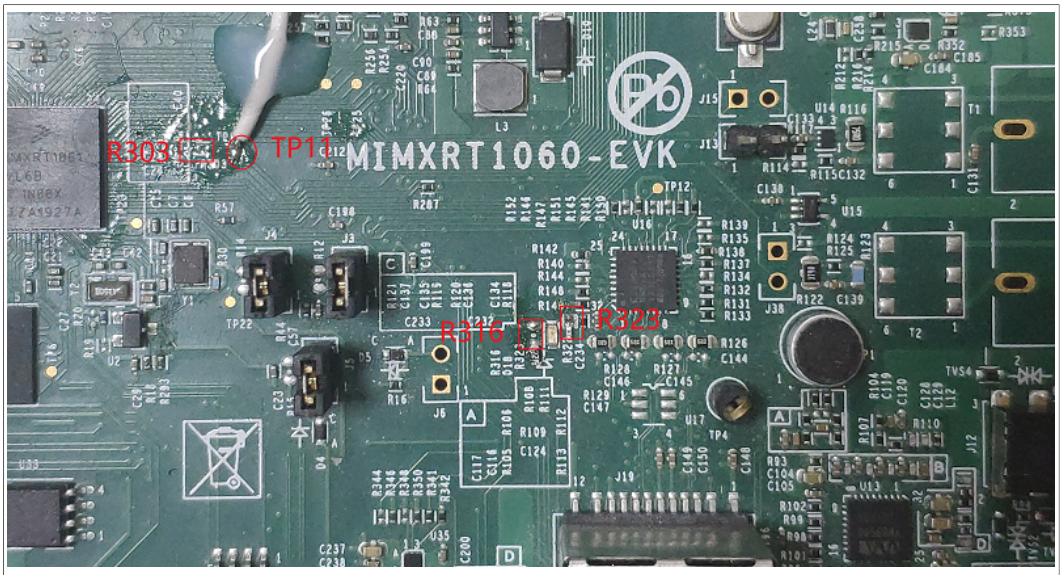


Figure 33. MIMXRT1060-EVK

Note:

To support HFP feature, you must remove R316 and R323 on MIMXRT1060-EVK.
To run HFP feature, you must remove J47 jumper.

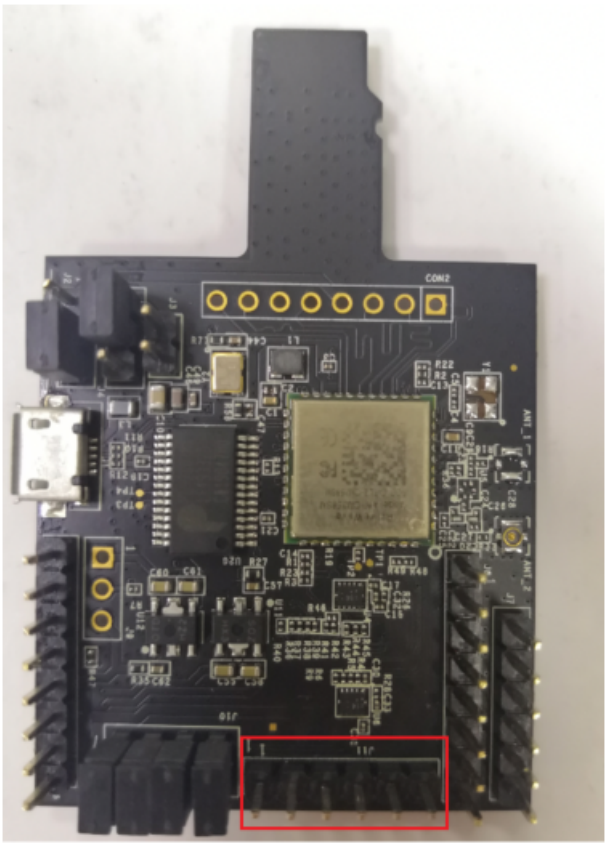


Figure 34. AW-CM358-uSD

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

9 Hardware Rework Guide for MIMXRT1060-EVK and Murata uSD-M.2 Adapter

This section is a brief hardware rework guidance of the Edgefast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVK board and Murata uSD-M.2 adapter. For details on Murata uSD-M.2 Adapter, see the Murata's uSD-M.2 webpage.The hardware rework consists only of HCI UART port rework.

9.1 Hardware rework

- **HCI UART rework**

Connect the TX/RX/RTS/CTS pins of two boards as the following table using jumper cables include in Murata's uSD-M.2 Adapter kit.

Table 13. Connect HCI UART pins

Pin name	uSD-M.2 adapter pin	i.MX RT1060 pin	Pin name of RT1060	GPIO name of RT1060
BT_UART_TXD_ HOST	J9 (pin 1)	J22 (pin 1)	LPUART3_RXD	GPIO_AD_B1_07
BT_UART_RXD_ HOST	J9 (pin 2)	J22 (pin 2)	LPUART3_TXD	GPIO_AD_B1_06
BT_UART_RTS_ HOST	J8 (pin 3)	J23 (pin 3)	LPUART3_CTS	GPIO_AD_B1_04
BT_UART_CTS_ HOST	J8 (pin 4)	J23 (pin 4)	LPUART3_RTS	GPIO_AD_B1_05

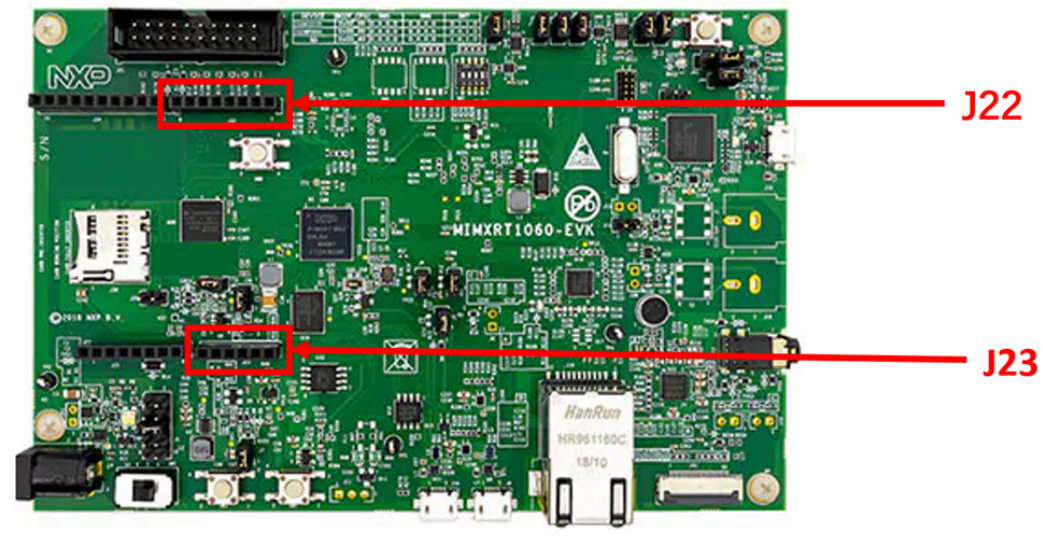


Figure 35. MIMXRT1060-EVK

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

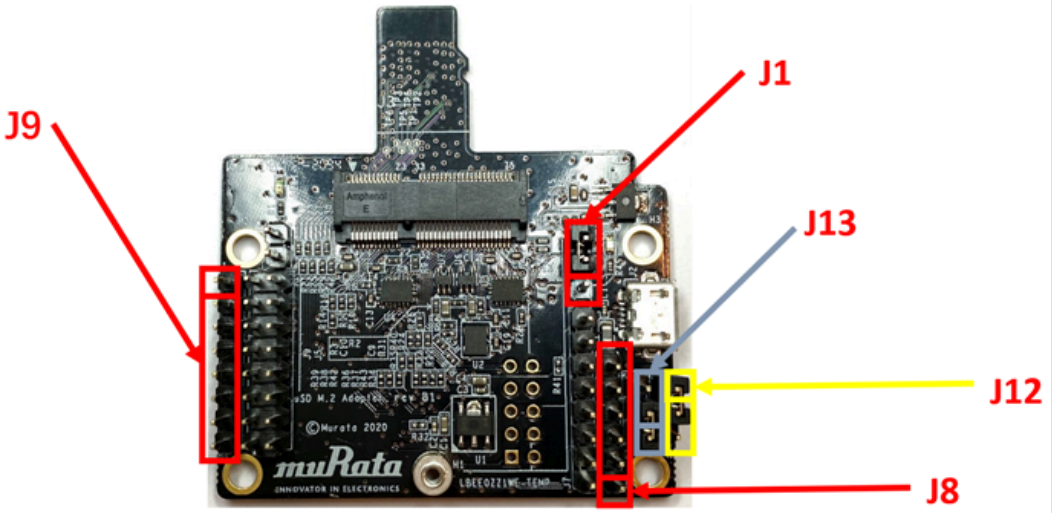


Figure 36. Murata uSD-M.2 adapter

Murata uSD-M.2 jumper settings:

- Both J12 and J13 = 1-2 (WLAN-SDIO = 1.8 V; and BT-UART and WLAN/BT-CTRL = 3.3 V)
- J1 = 2-3 (3.3 V from uSD connector)

10 Hardware Rework Guide for MIMXRT1060-EVK and AW-AM510-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1060-EVK board and AW-AM510-uSD. The AW-AM510-uSD user guide is available here. The hardware rework has one part:

- HCI UART port rework

10.1 Hardware rework

- **HCI UART rework**
  Make sure resistors R368/R376/R347/R349/R365/R363/R193/R186 are removed. Connect the pins of two boards as the following table.

Table 14. Connect pins

Pin Name	AW-AM510-uSD	i.MXRT1060	PIN NAME	GPIO NAME of RT1060
UART_TXD	J10 (pin 4)	J22 (pin 1)	LPUART3_RXD	GPIO_AD_B1_07
UART_RXD	J10 (pin 2)	J22 (pin 2)	LPUART3_TXD	GPIO_AD_B1_06
UART_RTS	J10 (pin 6)	J23 (pin 3)	LPUART3_CTS	GPIO_AD_B1_04
UART_CTS	J10 (pin 8)	J23 (pin 4)	LPUART3_RTS	GPIO_AD_B1_05
GND	J6 (pin 7)	J25 (pin 7)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

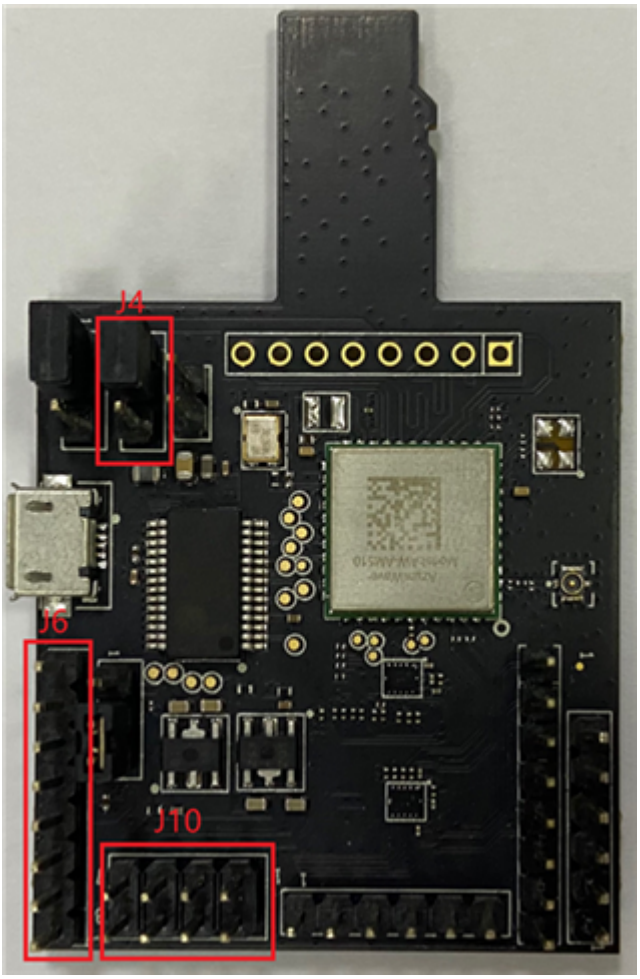


Figure 37. AW-AM510-uSD

Jumper Setting:

– Connect J4[2-3] for VIO 3.3 V supply

- **PCM interface rework**
  Connect the pins of two boards as the following table.

Table 15. Connect pins

PIN NAME	AW-AM510-USD	I.MXRT1060	PIN NAME OF RT1060	GPIO NAME of RT1060
PCM_IN	J11 (pin 1)	J22 (pin 5)	SAI2_TXD	GPIO_AD_B0_09
PCM_OUT	J11 (pin 2)	TP11	SAI2_RXD	GPIO_AD_B0_08
PCM_SYNC	J11 (pin 3)	J21 (pin 9)	SAI2_RX_SYNC	>GPIO_AD_01_07
PCM_CLK	J11 (pin 4)	J21 (pin 7)	SAI2_RX_BCLK	GPIO_AD_B0_06
GND	J11 (pin 6)	J21 (pin 20)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

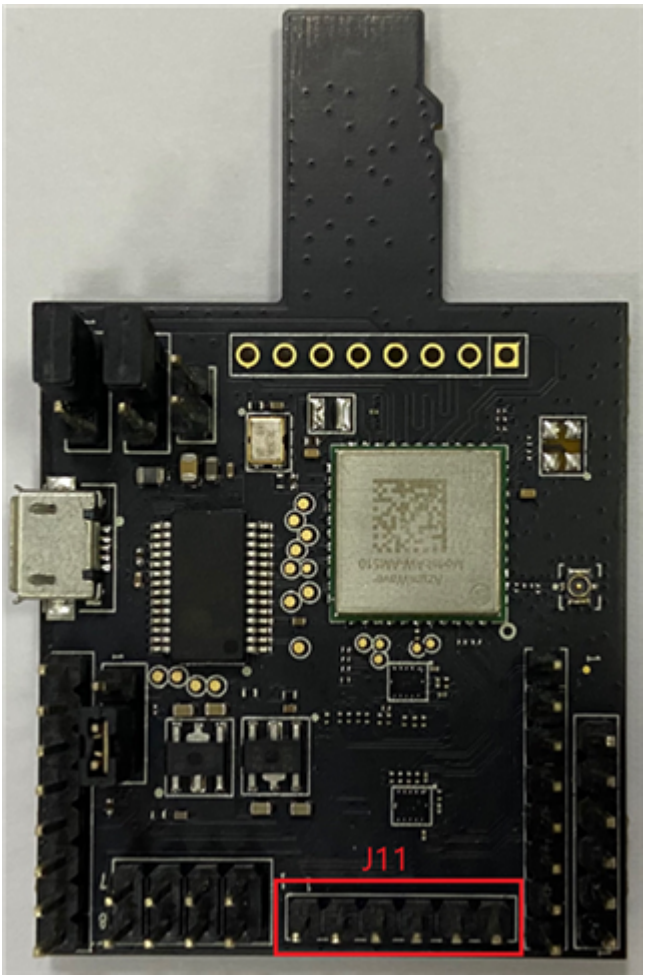


Figure 38. AW-AM510-uSD

11 Hardware Rework Guide for MIMXRT1170-EVK and AW-AM457-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1170-EVK board and AW-AM457-uSD. The AW-AM457-uSD user guide is available here. The hardware rework has one part:

- HCI UART port rework

11.1 Hardware rework

1. HCI UART rework
   Connect the pins of two boards as the following table.

Table 16. Connect pins

Pin Name	AW-AM457-uSD	i.MXRT1170	PIN NAME	GPIONAME of RT1170
UART_TXD	J10 (pin 4)	J25 (pin 13)	LPUART7_RXD	GPIO_AD_01
UART_RXD	J10 (pin 2)	J25 (pin 15)	LPUART7_TXD	GPIO_AD_00

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 16. Connect pins*...continued*

Pin Name	AW-AM457-uSD	i.MXRT1170	PIN NAME	GPIONAME of RT1170
UART_CTS	J10 (pin 8)	J25 (pin 9)	LPUART7_RTS	GPIO_AD_03
UART_RTS	J10 (pin 6)	J25 (pin 11)	LPUART7_CTS	GPIO_AD_02
GND	J6 (pin 7)	J26 (pin 1)	GND	GND

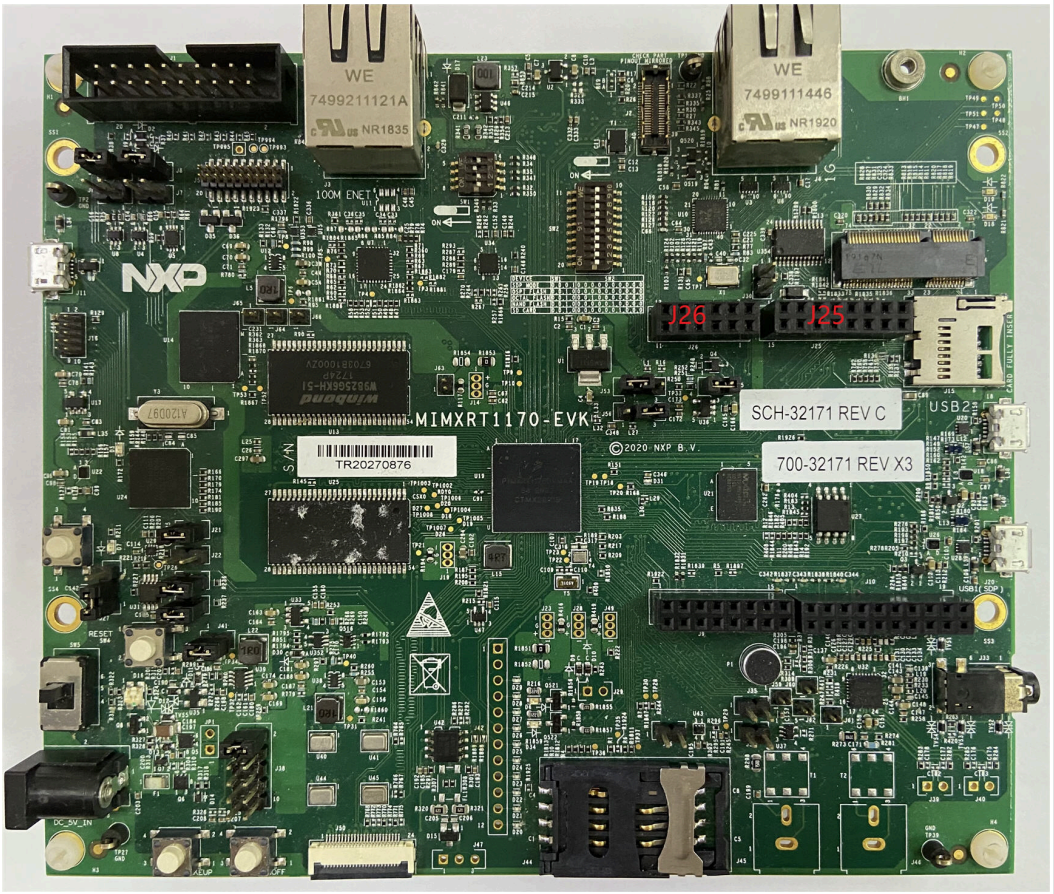


Figure 39. MIMXRT1170-EVK

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

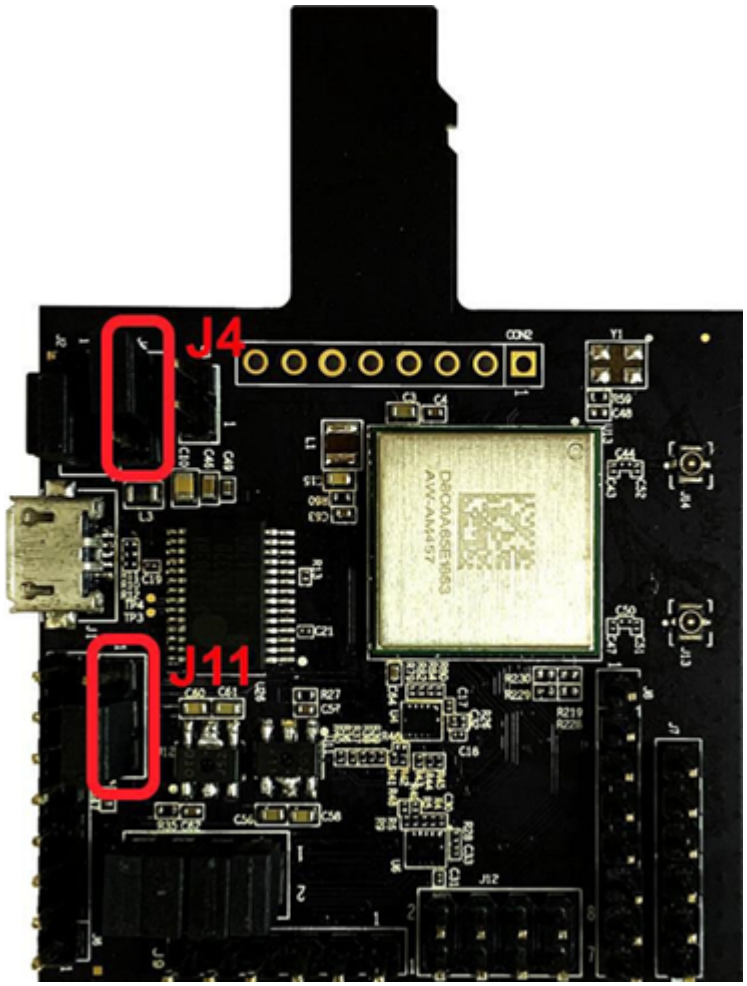


Figure 40. AW-AM457-uSD

Jumper Setting:

- Connect J4[2-3] for VIO 3.3 V supply
- Connect J11[2-3] for VIO_SD 3.3 V supply

12 Hardware Rework Guide for MIMXRT1170-EVK and AW-CM358MA

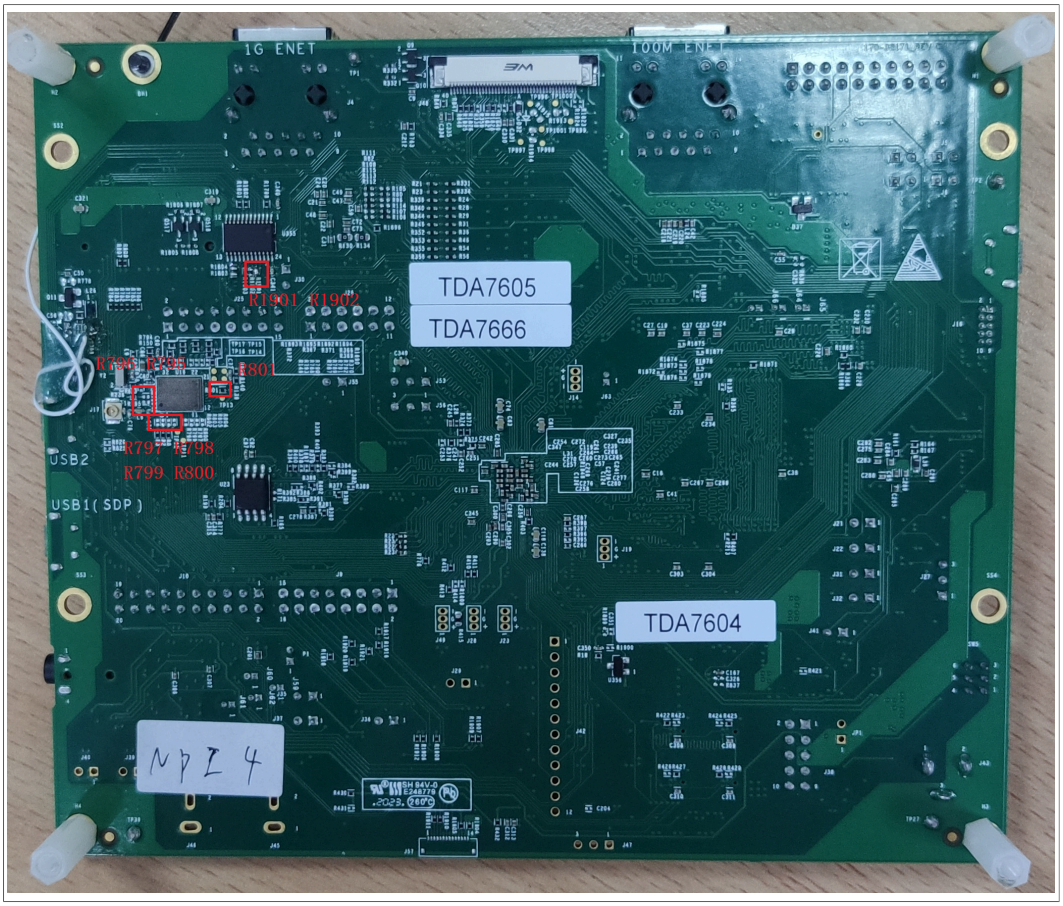
This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1170-EVK board and AW-CM358MA. The User's Guide of AW-CM358MA is available here.

12.1 Hardware rework

- **HCI UART rework**
  1. Solder 0 ohm resistor to R1901 and R1902.
  2. Remove resistors R795, R796, R797, R798, R799, R800, and R801.

All information provided in this document is subject to legal disclaimers.

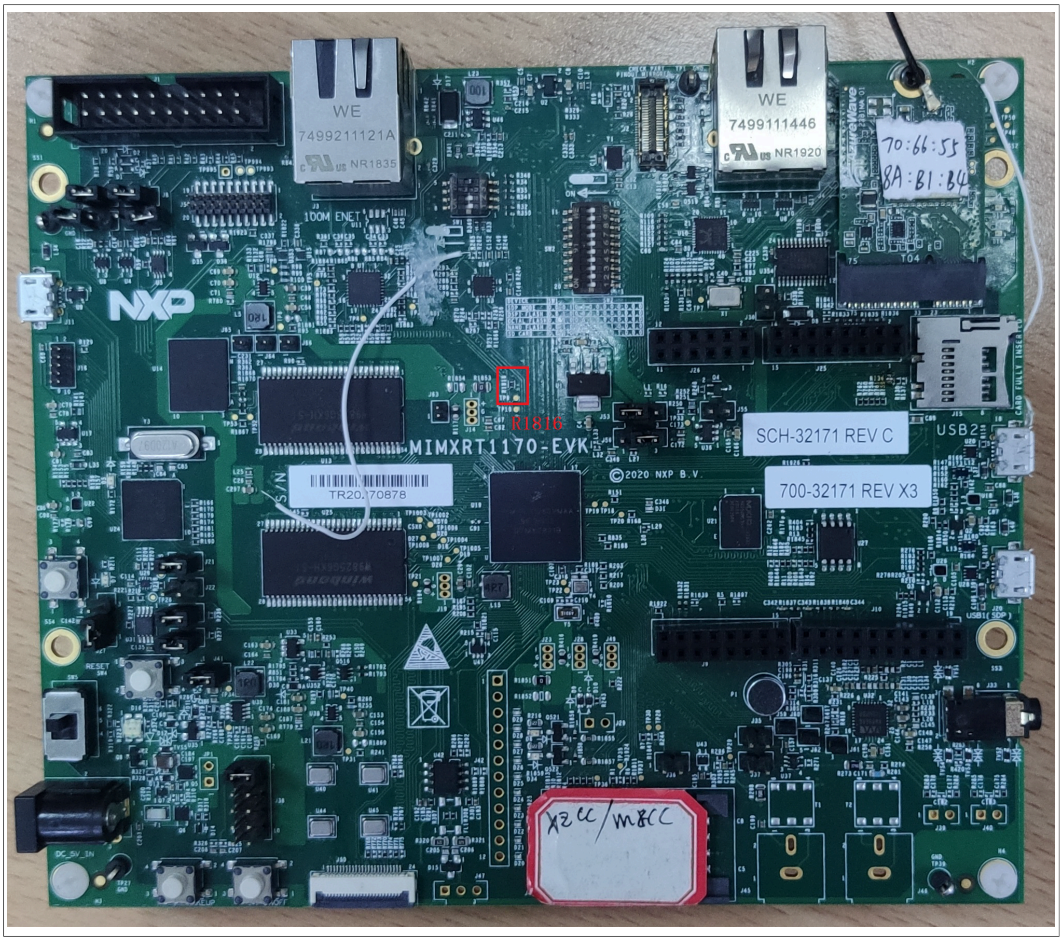
© 2022 NXP B.V. All rights reserved.



3. Remove 0 ohm resistor from R1816.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.



13 Hardware Rework Guide for MIMXRT1170-EVK and AW-AM510MA

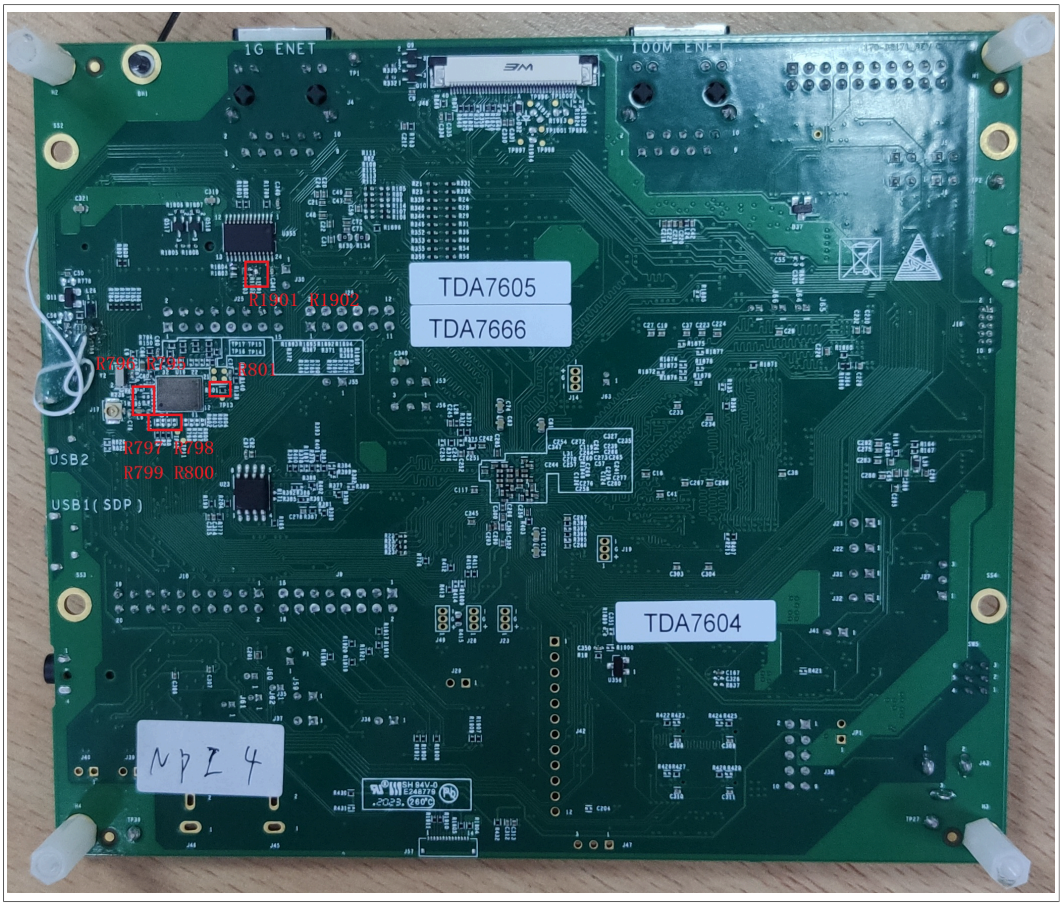
This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1170-EVK board and AW-AM510MA. The User's Guide of AW-AM510MA is available here.

13.1 Hardware rework

- **HCI UART rework**
  1. Solder 0 ohm resistor to R1901 and R1902.
  2. Remove resistors R795, R796, R797, R798, R799, R800, and R801.

All information provided in this document is subject to legal disclaimers.

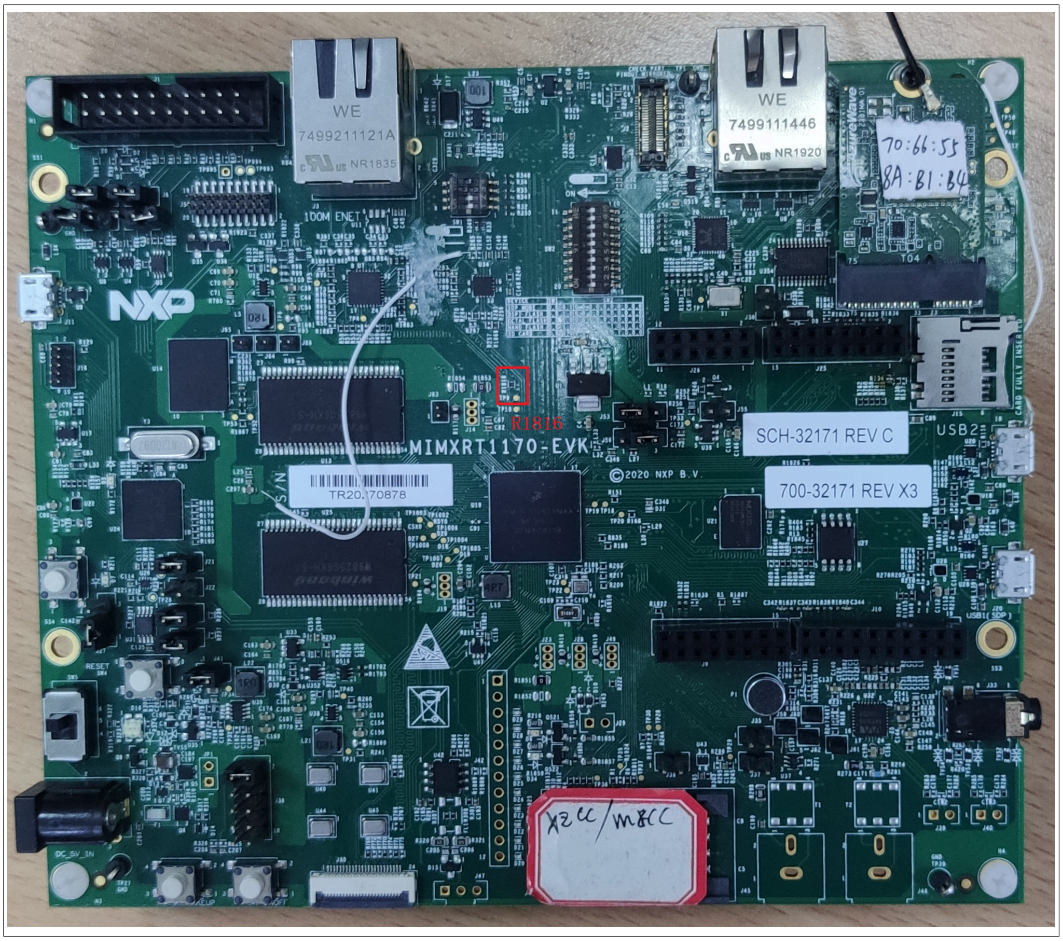
© 2022 NXP B.V. All rights reserved.



3. Remove 0 ohm resistor from R1816.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.



14 Hardware Rework Guide for MIMXRT1170-EVK with Direct Murata M.2 Module

This section is a brief hardware rework guidance of the Edgefast Bluetooth PAL on the NXP i.MX MIMXRT1170-EVK board and the Murata 1XK or 1ZM solution - direct M.2 connection to Embedded Artists' EAR00385 (1XK) or EAR00364 (1ZM) M.2 modules. The rework is necessary to connect the RADIO_ENABLE (or WL_RTS) signal in addition to two UART signals.

The hardware rework consists of two parts:

- HCI UART port rework
- Radio control signal rework (RADIO_ENABLE)

14.1 Hardware rework

1. **HCI UART rework:**
   On MIMXRT1170-EVK board, populate R1901 and R1902 with zero Ohm resistors. This connects the BT_UART_RXD and BT_UART_CTS signals respectively to the M.2 connector.
2. **Radio control signal rework (WL_CTS):**

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

On MIMXRT1170-EVK board, populate R404 with a zero Ohm resistor. Remove R183 resistor. This connects the WL_RST signal directly to the M.2 connector. This signal resets both wireless cores.

15 Hardware Rework Guide for MIMXRT1170-EVK and Murata uSD-M.2 Adapter

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1170-EVK board and Murata uSD-M.2 adapter. For details on the Murata uSD-M.2 Adapter, see the Murata's uSD-M.2 webpage.

The hardware rework consists only of HCI UART port rework.

15.1 Hardware rework

HCI UART rework:

Connect the TX/RX/RTS/CTS pins of two boards as the following table using jumper cables include in the Murata's uSD-M.2 Adapter kit.

Table 17. Connect HCI UART pins

Pin name	uSD-M.2 adapter pin	i.MX RT1170 pin	Pin name	GPIO name of RT1170
BT_UART_TXD_HOST	J9 (pin 1)	J25 (pin 13)	LPUART7_RXD	GPIO_AD_01
BT_UART_RXD_HOST	J9 (pin 2)	J25 (pin 15)	LPUART7_TXD	GPIO_AD_00
BT_UART_CTS_HOST	J8 (pin 4)	J25 (pin 9)	LPUART7_RTS	GPIO_AD_03
BT_UART_RTS_HOST	J8 (pin 3)	J25 (pin 11)	LPUART7_CTS	GPIO_AD_02

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

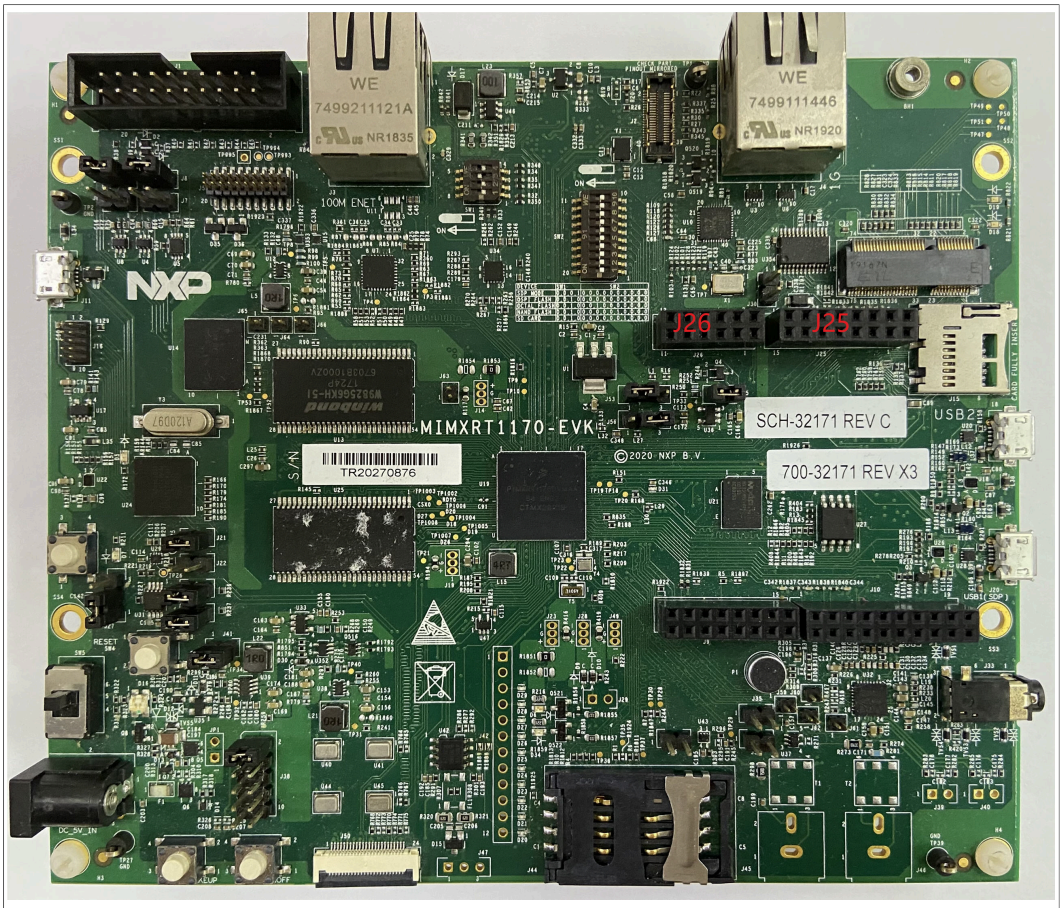


Figure 41. MIMXRT1170-EVK

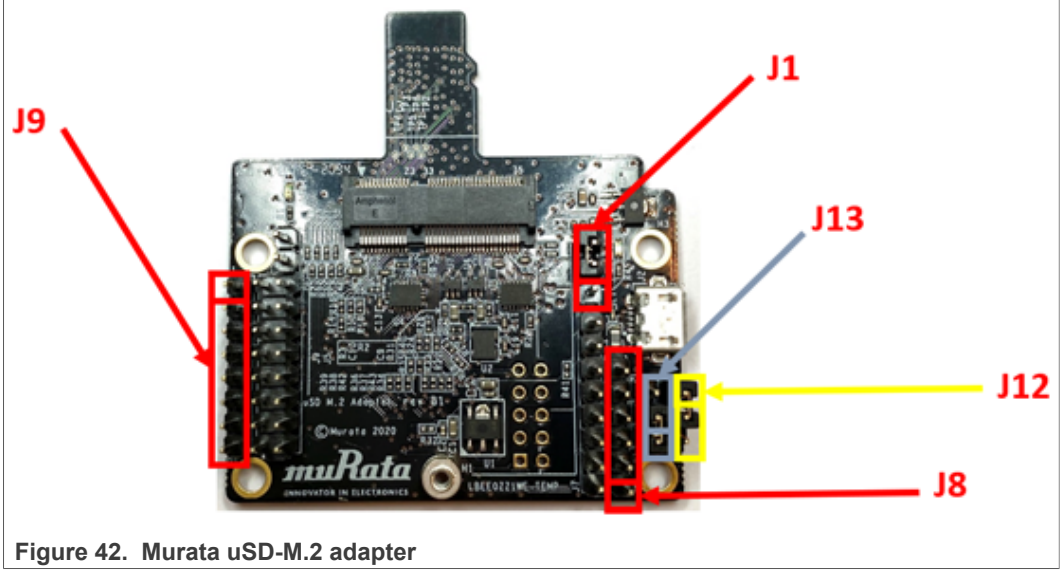


Figure 42. Murata uSD-M.2 adapter

Murata uSD-M.2 jumper settings:

- Both J12 and J13 = 1-2 (WLAN-SDIO = 1.8 V; and BT-UART and WLAN/BT-CTRL = 3.3 V)
- J1 = 2-3 (3.3 V from uSD connector)

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

16 Hardware Rework Guide for MIMXRT1170-EVKB and Murata 1XK M.2 Adapter

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1170-EVKB and the Murata 1XK M.2 solution - direct M.2 connection to Embedded Artists' EAR00385 (1XK) M.2 modules.

The hardware rework has two parts:

- HCI UART rework
- PCM interface rework

16.1 Hardware rework

- **HCI UART rework**
  1. Remove resistors R183 and R1816.
  2. Solder 0 ohm resistor to R404, R1901, and R1902.

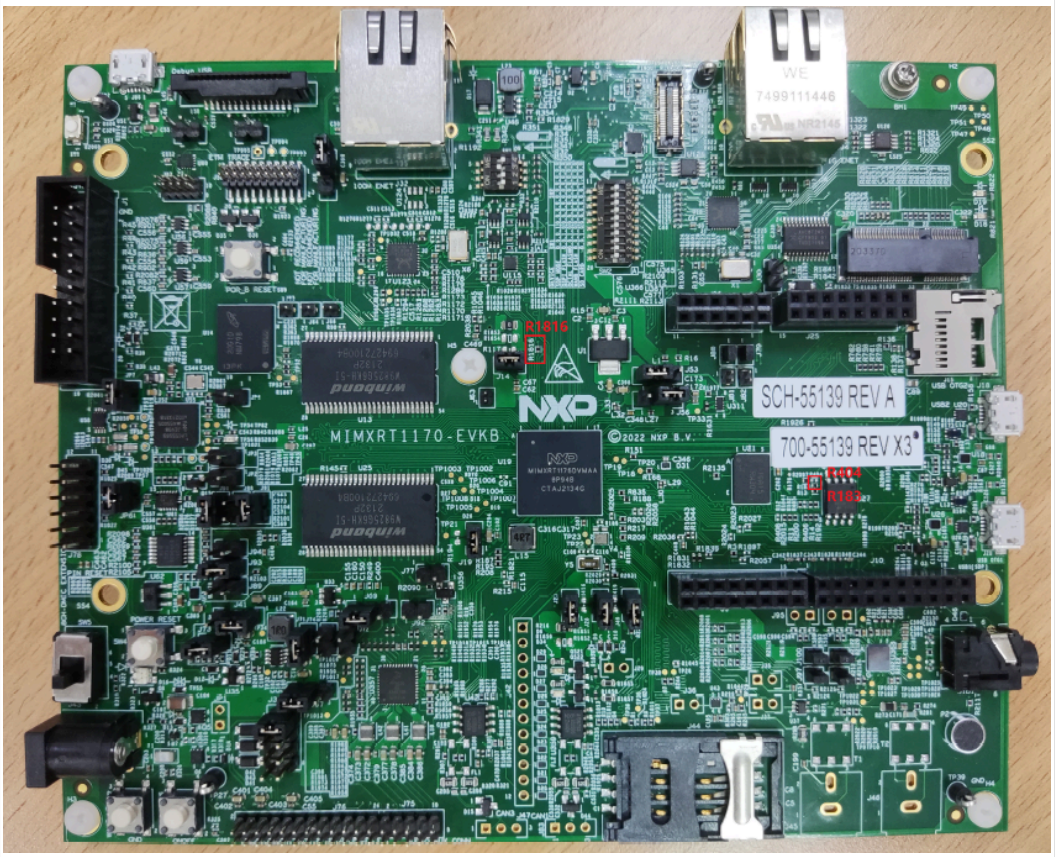


Figure 43. MIMXRT1170-EVKB UART Front

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

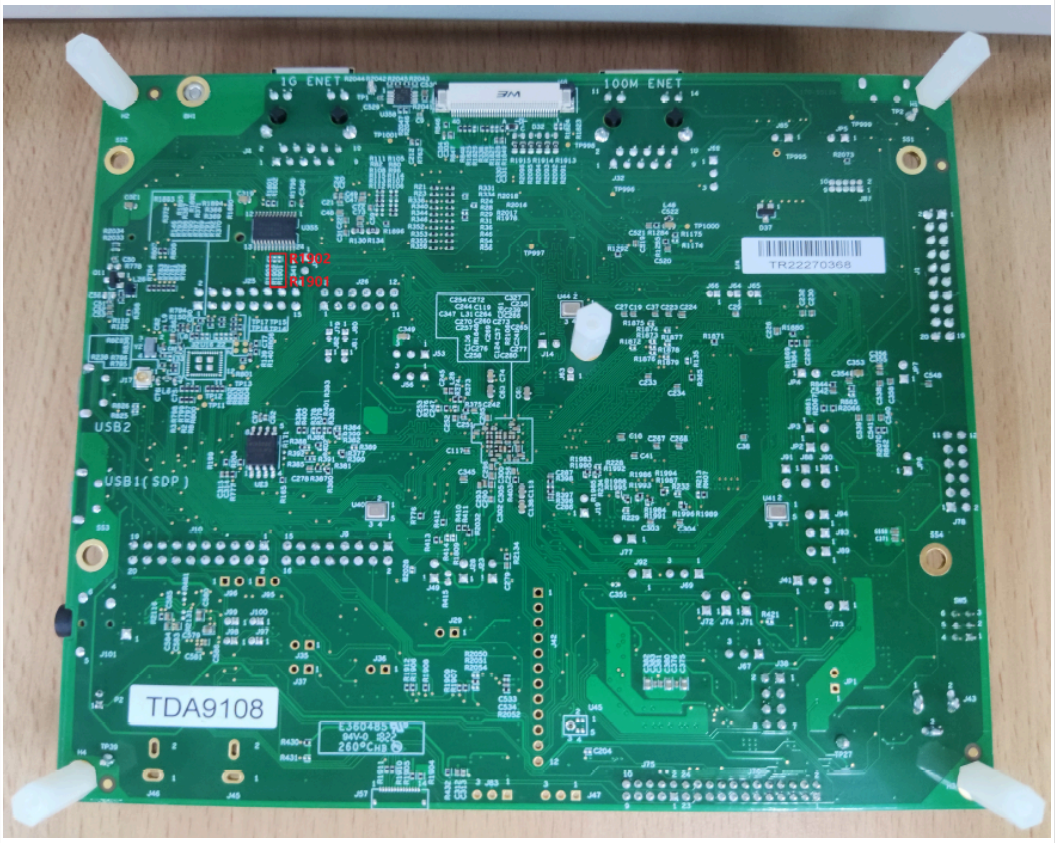


Figure 44. MIMXRT1170-EVKB UART Back

- **PCM interface rework**
  1. Disconnect header J79 and J80.
  2. Connect header J81 and J82.
  3. Remove resistors R1985, R1986, R1987, R1988, R1992, R1993, R1994, and R1995.
  4. Solder 0 ohm resistor to R228, R229, R232, R234, and R1903.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

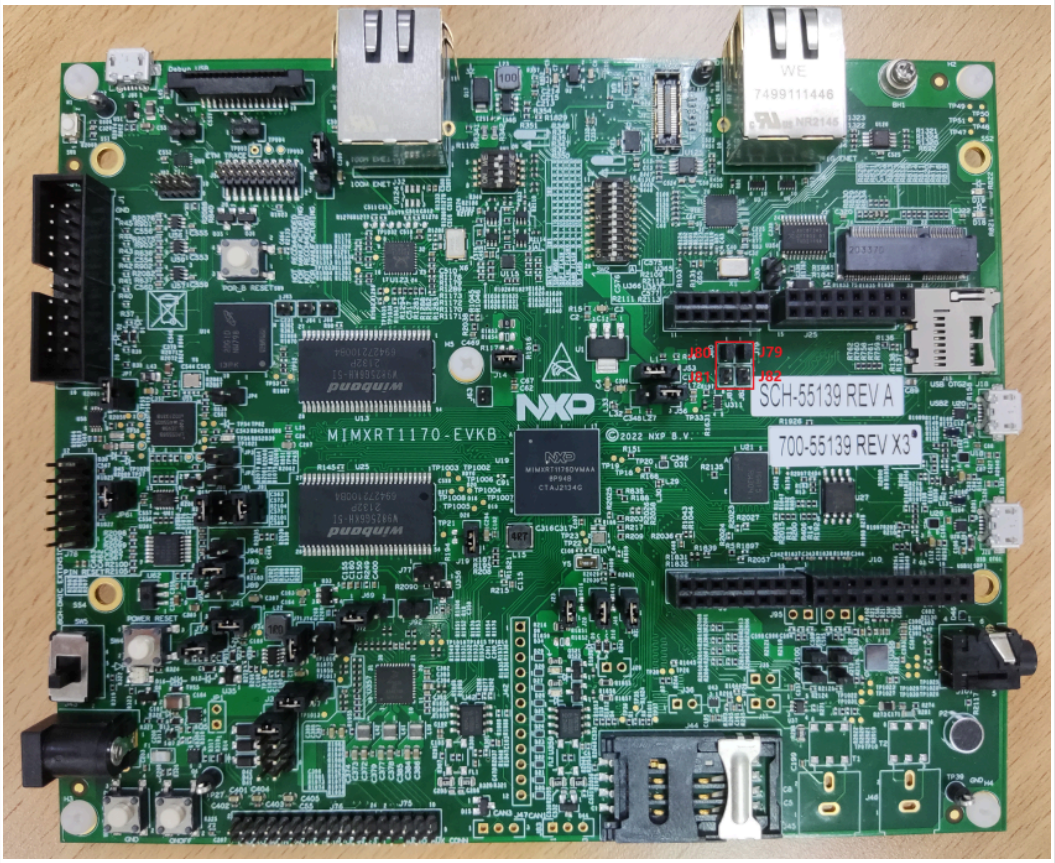


Figure 45. MIMXRT1170-EVKB PCM Front

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

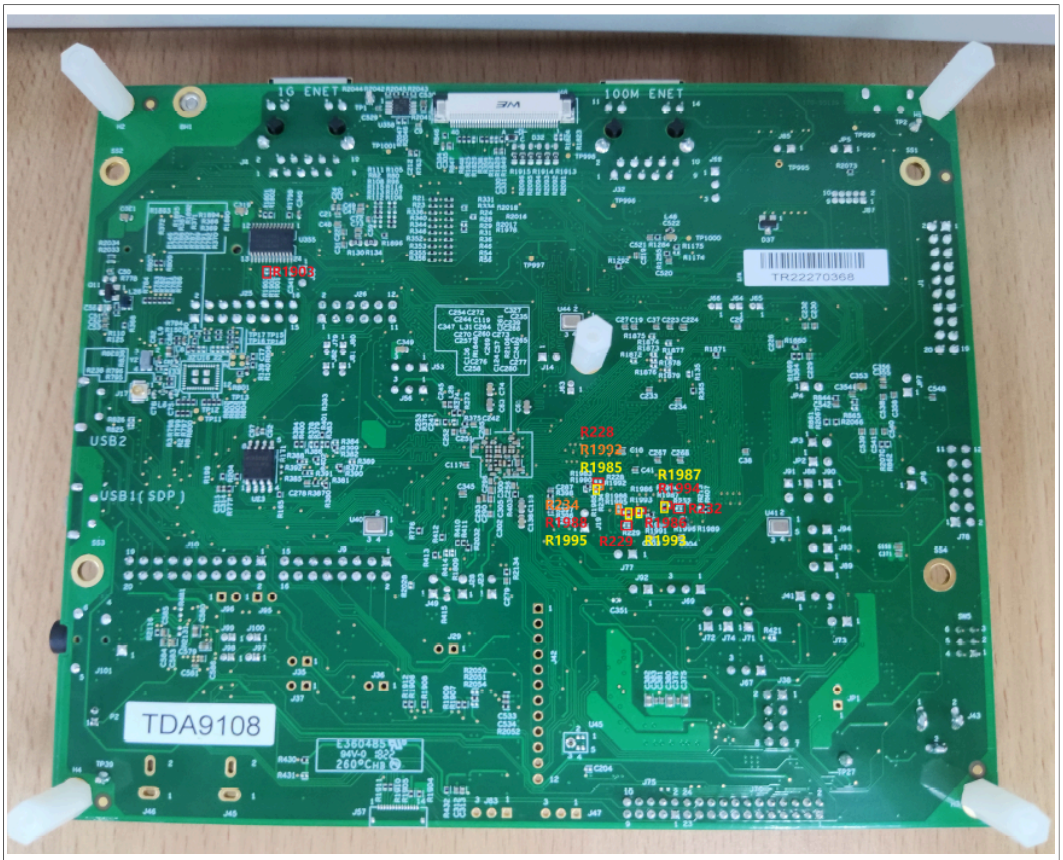


Figure 46. MIMXRT1170-EVKB PCM Back

17 Hardware Rework Guide for MIMXRT1170-EVKB and Murata 1ZM M.2 Adapter

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT1170-EVKB and the Murata 1ZM M.2 solution - direct M.2 connection to Embedded Artists' EAR00364 (1ZM) M.2 modules.

The hardware rework has two parts:

- HCI UART rework
- PCM interface rework

17.1 Hardware rework

- **HCI UART rework**
  1. Remove resistors R183 and R1816.
  2. Solder 0 ohm resistor to R404, R1901, and R1902.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

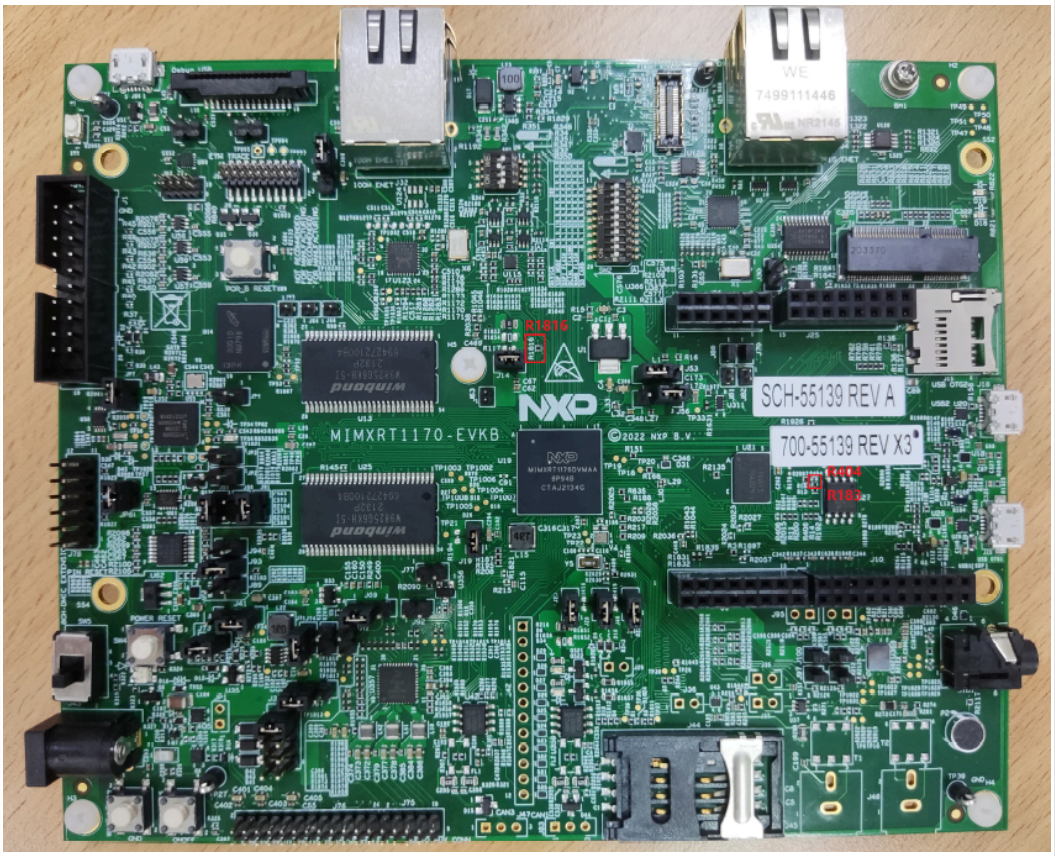


Figure 47. MIMXRT1170-EVKB UART Front

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

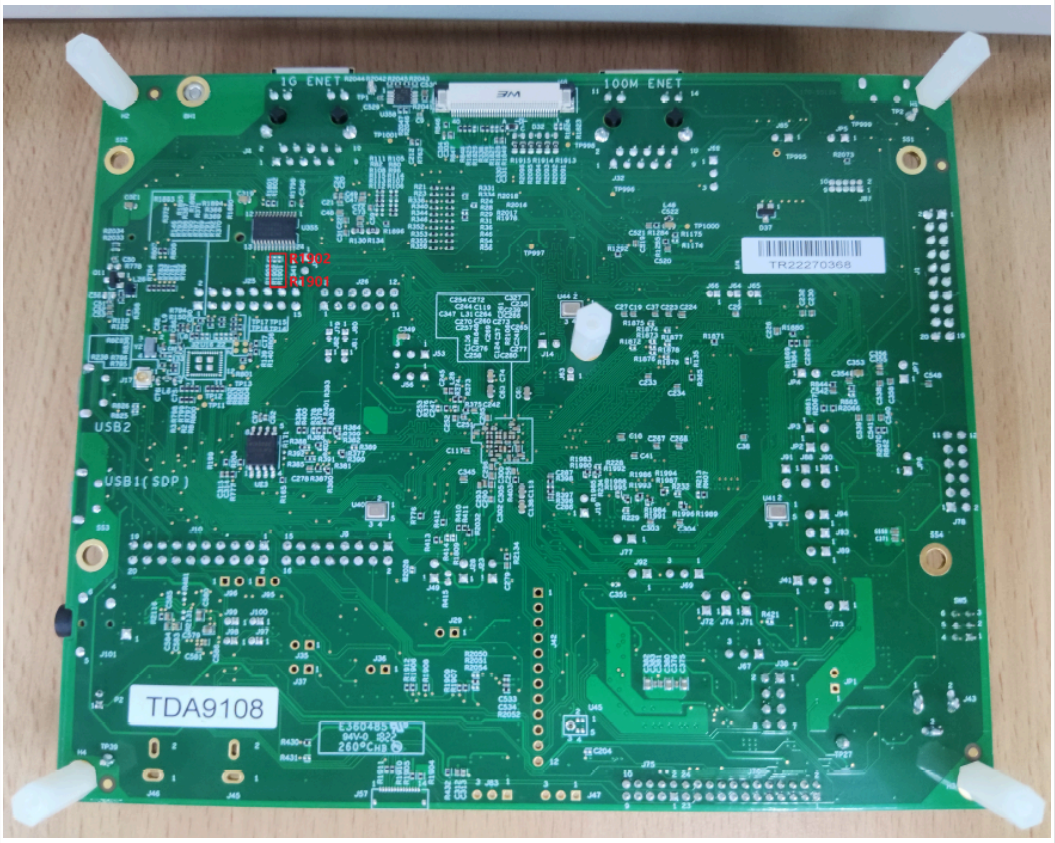


Figure 48. MIMXRT1170-EVKB UART Back

- **PCM interface rework**
  1. Disconnect header J79 and J80.
  2. Connect header J81 and J82.
  3. Remove resistors R1985, R1986, R1987, R1988, R1992, R1993, R1994, and R1995.
  4. Solder 0 ohm resistor to R228, R229, R232, R234, and R1903.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

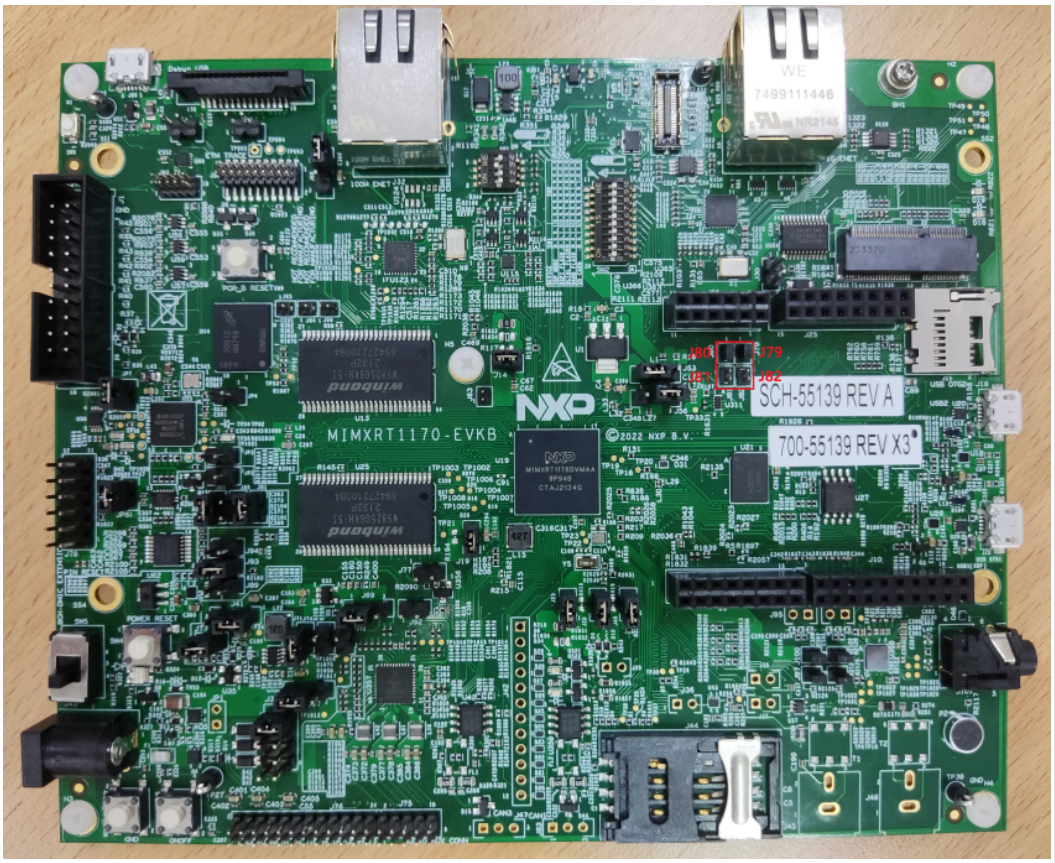


Figure 49. MIMXRT1170-EVKB PCM Front

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

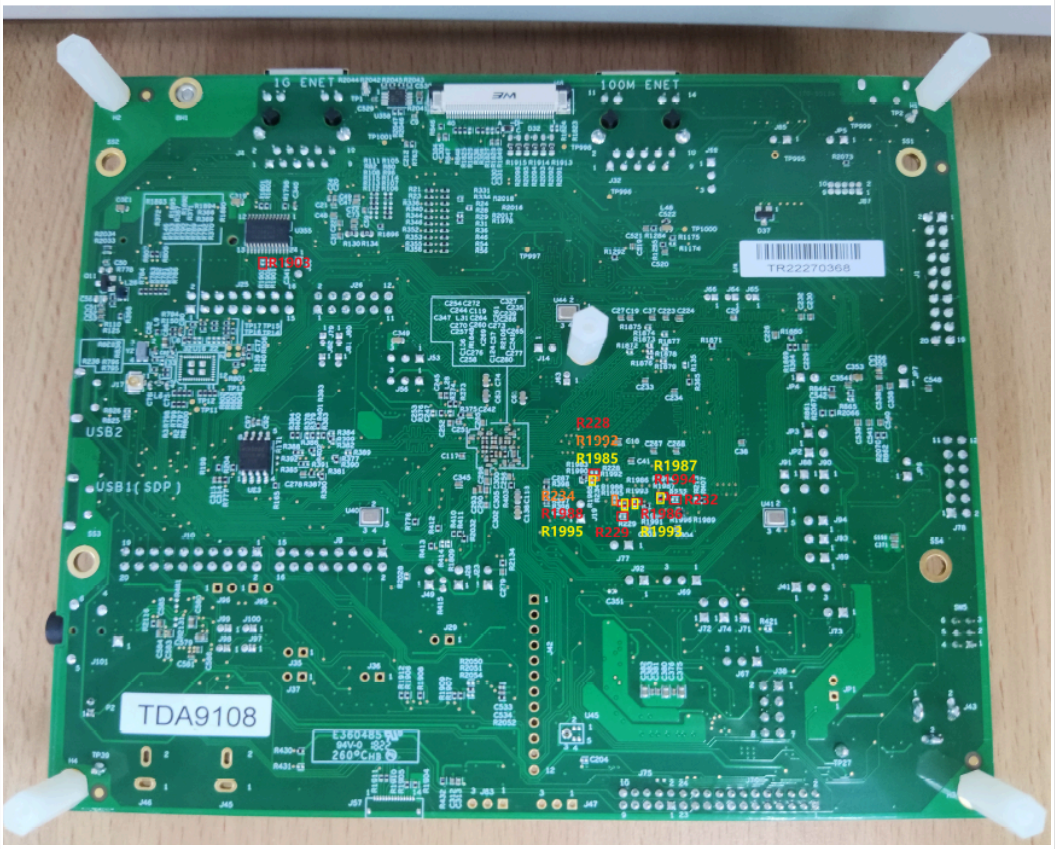


Figure 50. MIMXRT1170-EVKB PCM Back

18 Hardware Rework Guide for MIMXRT685-EVK and AW-AM457-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT685-EVK board and AW-AM457-uSD. The AW-AM457-uSD user guide is available here.The hardware rework has one part:

- HCI UART port rework

18.1 Hardware rework

HCI UART rework

- R398 move from 1-2 to 2-3
- JP12 2-3
- Connect the pins of two boards as the following table.

Table 18. Connect pins

Pin Name	AW-AM457-uSD	i.MX RT685	PIN NAME	GPIONAME of i.MX RT685
UART_TXD	J10 (pin 4)	J27 (pin 1)	USART4_RXD	FC4_RXD_SDA_MOSI_DATA
UART_RXD	J10 (pin 2)	J27 (pin 2)	USART4_TXD	FC4_TXD_SCL_MISO_WS
UART_RTS	J10 (pin 6)	J47 (pin 9)	USART4_CTS	FC4_CTS_SDA_SSEL0
UART_CTS	J10 (pin 8)	J27 (pin 5)	USART4_RTS	FC4_RTS_SCL_SSEL1

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 18. Connect pins*...continued*

Pin Name	AW-AM457-u SD	i.MX RT685	PIN NAME	GPIONAME of i.MX RT685
GND	J6 (pin 7)	J29 (pin 6)	GND	GND

Figure 51. MIMXRT685-EVK

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

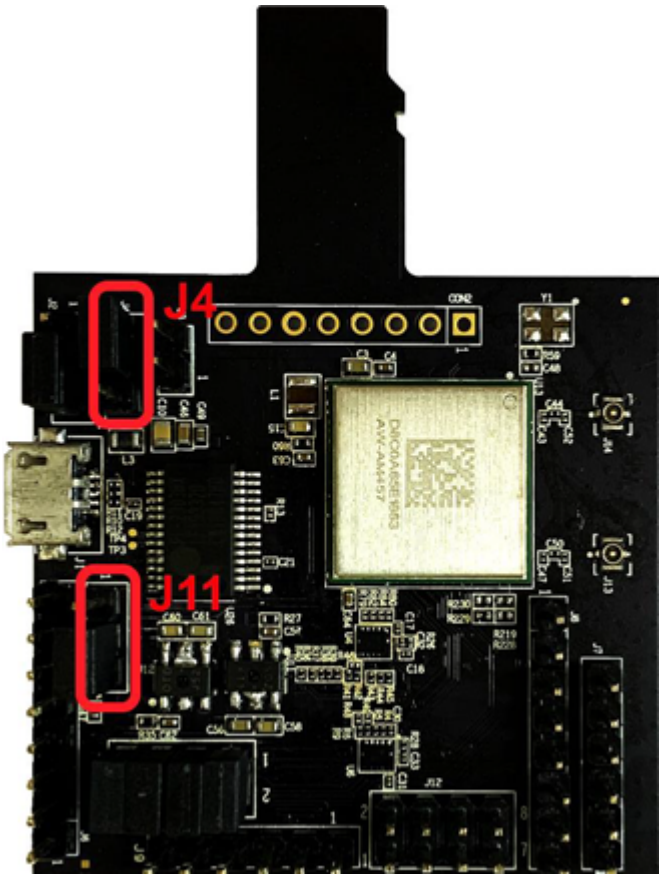


Figure 52. AW-AM457-uSD

Jumper Settings:

- Connect J4[2-3] for VIO 3.3 V supply
- Connect J11[2-3] for VIO_SD 3.3 V supply

PCM interface rework

Connect the pins of two boards as the following table.

Table 19. Connect pins

Pin Name	AW-AM457-uSD	i.MX RT685	PIN NAME of I.MX RT685	GPIONAME of I.MX RT685
PCM_IN	J9 (pin 1)	J47 (pin 7)	I2S2_TXD	FC2_RXD_SDA_MOSI_DATA
PCM_OUT	J9 (pin 2)	J28 (pin 4)	I2S5_RXD	FC5_RXD_SDA_MOSI_DATA
PCM_SYNC	J9 (pin 3)	J28 (pin 5)	I2S5_WS	FC5_TXD_SCL_MISO_WS
PCM_CLK	J9 (pin 4)	J28 (pin 6)	I2S5_SCK	FC5_SCK
GND	J9 (pin 6)	J29 (pin 7)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

19 Hardware Rework Guide for MIMXRT685-EVK and AW-CM358-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT685-EVK board and AW-CM358-uSD. The AW-CM358-uSD user guide is available here. The hardware rework has one part:

- HCI UART port rework

19.1 Hardware rework

HCI UART rework

R398 move from 1-2 to 2-3.

Connect the pins of two boards as the following table.

Table 20. Connect pins

Pin Name	AW-CM358-USD	i.MXRT685	PIN NAME	GPIONAME of RT685
UART_TXD	J10 (pin 4)	J27 (pin 1)	USART4_RXD	FC4_RXD_SDA_MOSI_DATA
UART_RXD	J10 (pin 2)	J27 (pin 2)	USART4_TXD	FC4_TXD_SCL_MISO_WS
UART_RTS	J10 (pin 6)	J47 (pin 9)	USART4_CTS	FC4_CTS_SDA_SSEL0
UART_CTS	J10 (pin 8)	J27 (pin 5)	USART4_RTS	FC4_RTS_SCL_SSEL1
GND	J6 (pin 7)	J29 (pin 6)	GND	GND

Figure 53. MIMXRT685-EVK

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

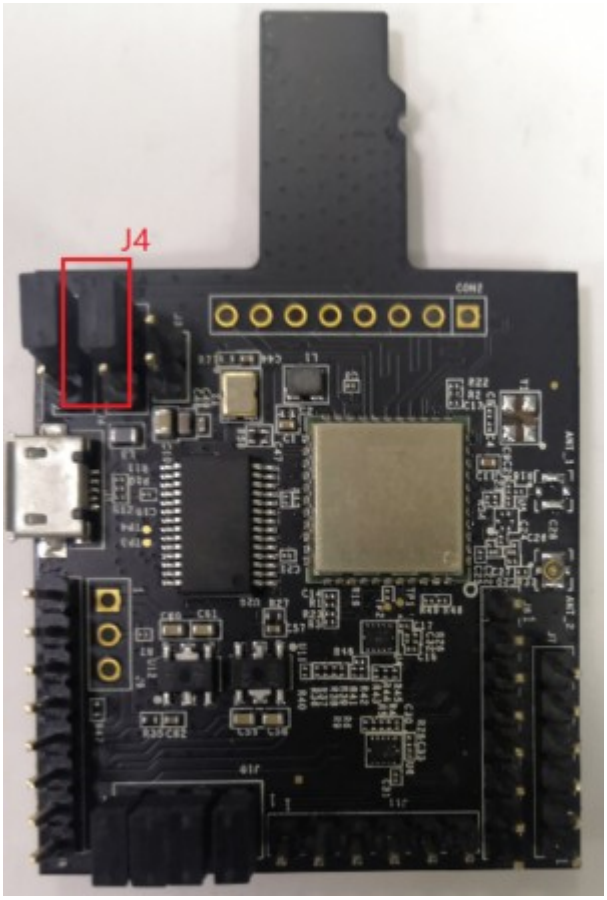


Figure 54. AW-CM358-uSD

Jumper Setting:

Connect J4[1-2] for VIO 1.8 V supply.

PCM interface rework

Connect the pins of two boards as the following table.

Table 21. Connect pins

Pin Name	AW-CM358-USD	i.MX RT685	PIN NAME of RT685	GPIONAME of RT685
PCM_IN	J11 (pin 1)	J47 (pin 7)	I2S2_TXD	FC2_RXD_SDA_MOSI_DATA
PCM_OUT	J11 (pin 2)	J28 (pin 4)	I2S5_RXD	FC5_RXD_SDA_MOSI_DATA
PCM_SYNC	J11 (pin 3)	J28 (pin 5)	I2S5_WS	FC5_TXD_SCL_MISO_WS
PCM_CLK	J11 (pin 4)	J28 (pin 6)	I2S5_SCK	FC5_SCK
GND	J11 (pin 5)	J29 (pin 7)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

20 Hardware Rework Guide for MIMXRT685-EVK and AW-AM510-uSD

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT685-EVK board and AW-AM510-uSD. The AW-AM510-uSD user guide is available here. The hardware rework has one part:

- HCI UART port rework

20.1 Hardware rework

- **HCI UART rework**
  Make sure resistors R368/R376/R347/R349/R365/R363/R193/R186 are removed. Connect the pins of two boards as the following table.

Table 22. Connect pins

Pin Name	AW-AM510-uSD	i.MXRT685	PIN NAME	GPIO NAME of RT685
UART_TXD	J10 (pin 4)	J27 (pin 1)	USART4_RXD	FC4_RXD_SDA_MOSI_DATA
UART_RXD	J10 (pin 2)	J27 (pin 2)	USART4_TXD	FC4_TXD_SCL_MISO_WS
UART_RTS	J10 (pin 6)	J47 (pin 9)	USART4_CTS	FC4_CTS_SDA_SSEL0
UART_CTS	J10 (pin 8)	J27 (pin 5)	USART4_RTS	FC4_RTS_SCL_SSEL1
GND	J6 (pin 7)	J29 (pin 6)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

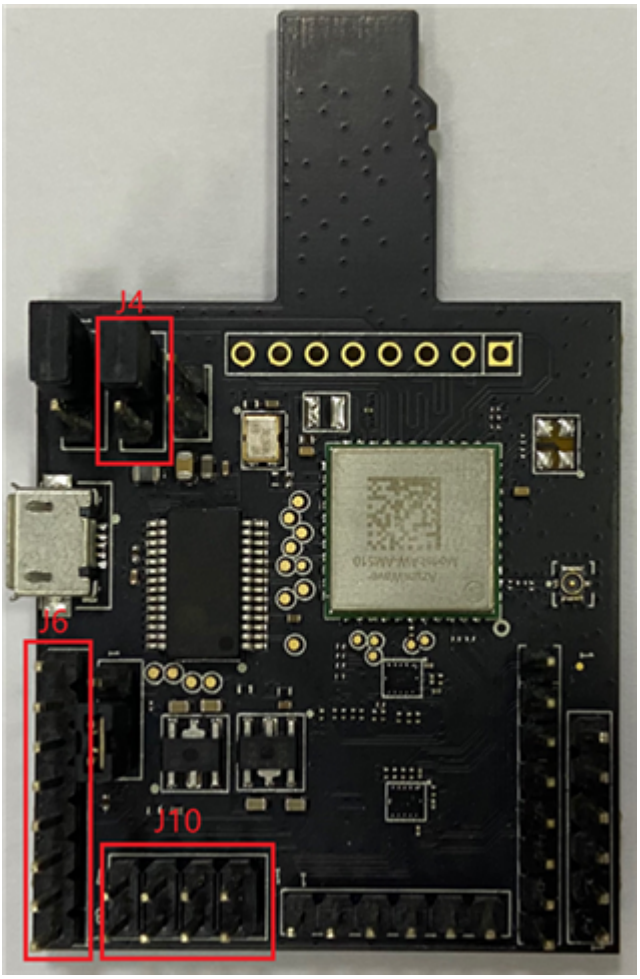


Figure 55. AW-AM510-uSD

Jumper Setting:

– Connect J4[2-3] for VIO 3.3 V supply

- **PCM interface rework**
  Connect the pins of two boards as the following table.

Table 23. Connect pins

PIN NAME	AW-AM510-USD	i.MX RT685	PIN NAME of RT685	GPIONAME of RT685
PCM_IN	J11 (pin 1)	J47 (pin 7)	I2S2_TXD	FC2_RXD_SDA_MOSI_DATA
PCM_OUT	J11 (pin 2)	J28 (pin 4)	I2S5_RXD	FC5_RXD_SDA_MOSI_DATA
PCM_SYNC	J11 (pin 3)	J28 (pin 5)	I2S5_WS	FC5_TXD_SCL_MISO_WS
PCM_CLK	J11 (pin 4)	J28 (pin 6)	I2S5_SCK	FC5_SCK
GND	J11 (pin 6)	J29 (pin 7)	GND	GND

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

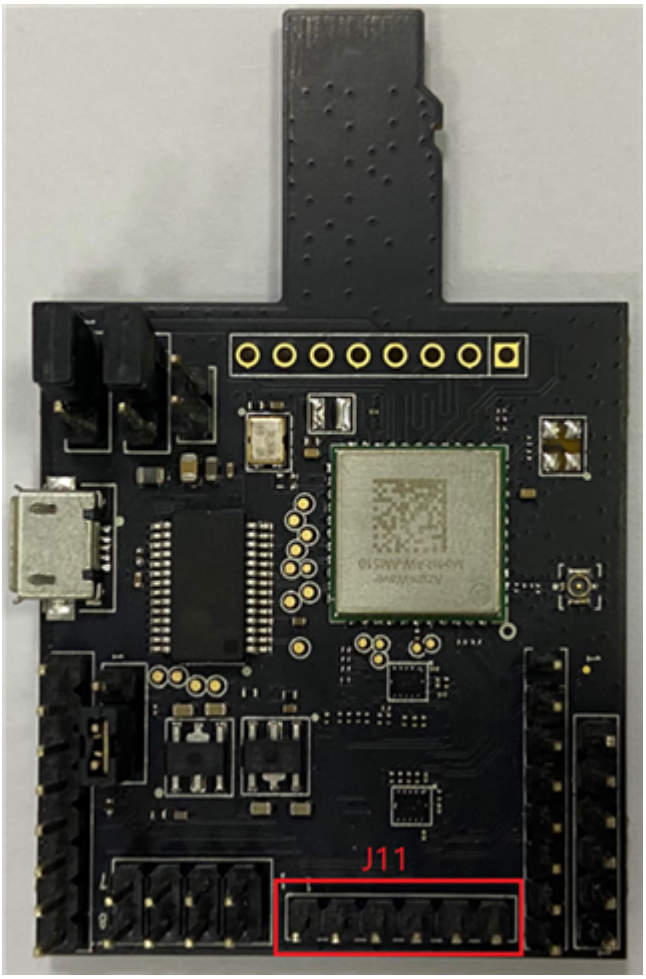


Figure 56. AW-AM510-uSD

21 Hardware Rework Guide for MIMXRT685-EVK and Murata uSD-M.2 Adapter

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT685-EVK board and the Murata uSD-M.2 adapter. For details on the Murata uSD-M.2 Adapter, see Murata's uSD-M.2 webpage.

The hardware rework has one part:

- HCI UART port rework

21.1 Hardware rework

HCI UART rework :

- JP12 2-3
- Connect the pins of two boards as the following table using jumper cables included in Murata's uSD-M.2 Adapter kit.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Table 24. Connect HCI UART pins

Pin name	uSD-M.2 adapter pin	i.MX RT685 pin	Pin name of RT685	GPIO name of RT685
BT_UART_TXD_ HOST	J9 (pin 1)	J27 (pin 1)	USART4_RXD	FC4_RXD_SDA_ MOSI_DATA
BT_UART_RXD_ HOST	J9 (pin 2)	J27 (pin 2)	USART4_TXD	FC4_TXD_SCL_ MISO_WS
BT_UART_RTS_ HOST	J8 (pin 3)	J47 (pin 9)	USART4_CTS	FC4_CTS_SDA_ SSEL0
BT_UART_CTS_ HOST	J8 (pin 4)	J27 (pin 5)	USART4_RTS	FC4_RTS_SCL_ SSEL1

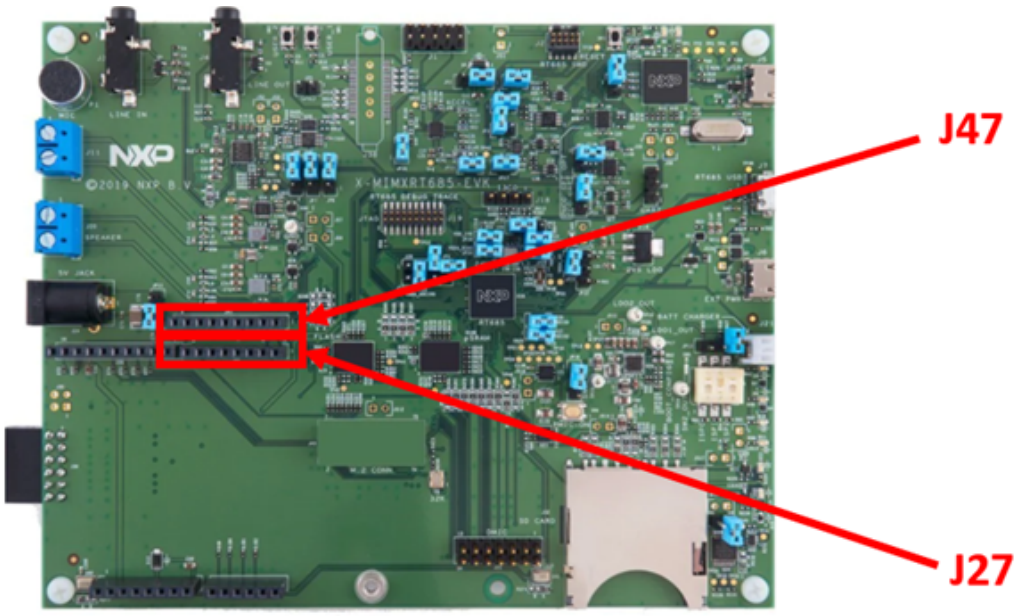


Figure 57. MIMXRT685-EVK

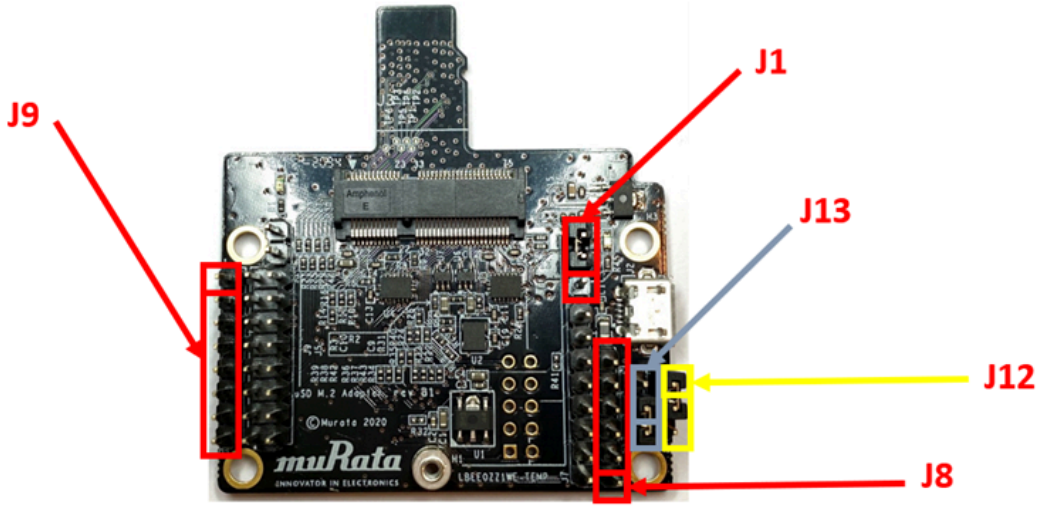


Figure 58. Murata uSD-M.2 adapter

Murata uSD-M.2 jumper settings:

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

- Both J12 and J13 = 1-2 (WLAN-SDIO = 1.8 V; and BT-UART and WLAN/BT-CTRL = 3.3 V)
- J1 = 2-3 (3.3 V from uSD connector)

22 Hardware Rework Guide for MIMXRT595-EVK with Direct Murata M.2 Module

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT595-EVK board and the Murata's 1XK or 1ZM solution - direct M.2 connection to Embedded Artists EAR00385 (1XK) or EAR00364 (1ZM) M.2 modules.

The hardware rework has one part:

- Debug console serial rework

22.1 Hardware rework

Debug console serial rework:

No special rework is required, except the following to enable the debug port.

- JP4 1-2.
- J27 1 - TX of USB to serial converter
- J27 2 - RX of USB to serial converter

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

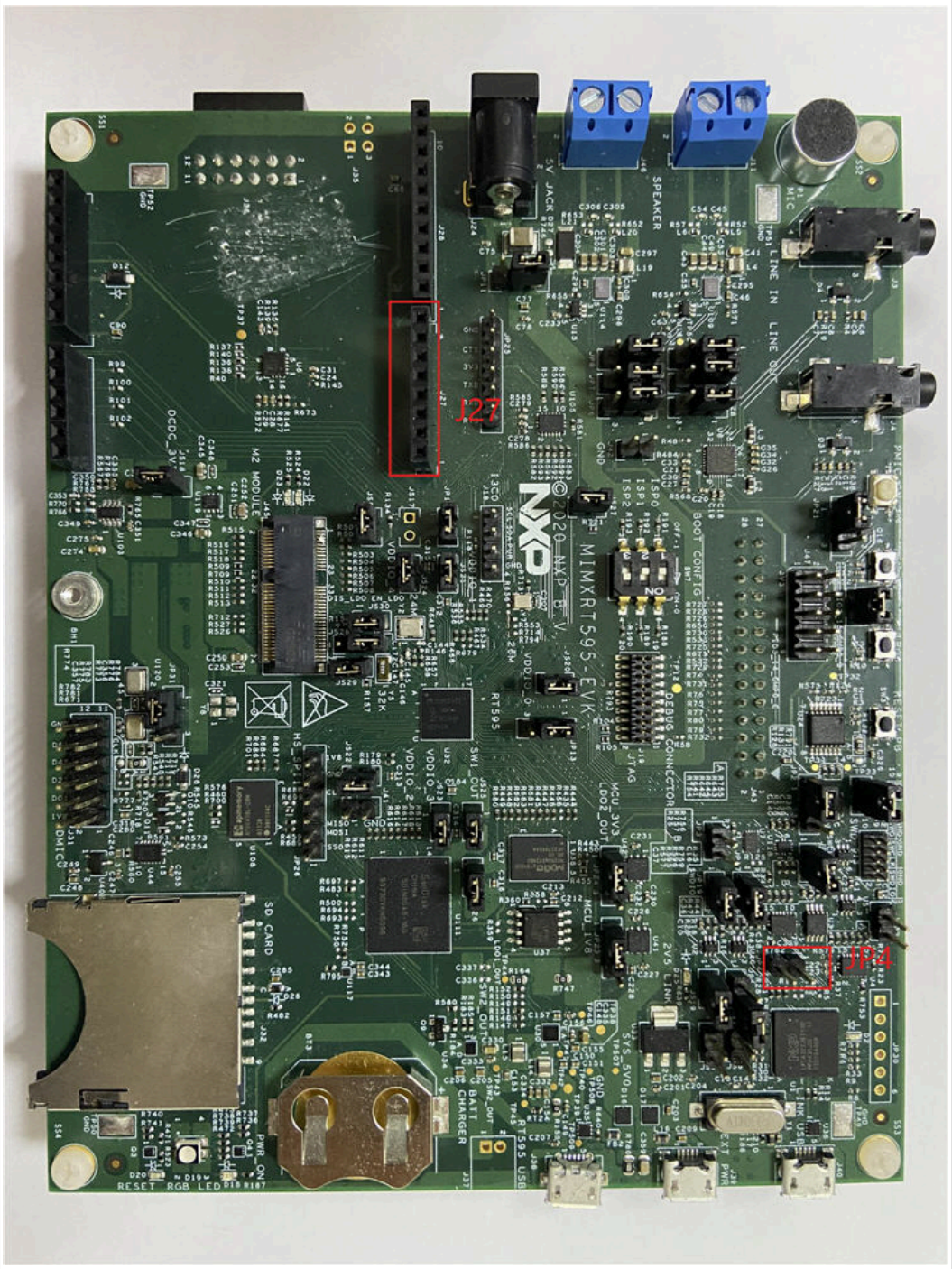


Figure 59. MIMXRT595

23 Hardware Rework Guide for MIMXRT595-EVK and AW-AM510MA

This section is a brief hardware rework guidance of the EdgeFast Bluetooth PAL on the NXP i.MX MIMXRT595-EVK board and AW-AM510MA. The AW-AM510MA user guide is available here. The hardware rework has one part:

- Debug console serial rework

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

23.1 Hardware rework

Debug console serial rework:

No special rework is required, except the following to enable the debug port.

- Connect J39 with external power.
- Connect JP4 1-2.
- J27 1 — TX of USB to serial converter.
- J27 2 — RX of USB to serial converter.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Figure 60. MIMXRT595-EVK

24 Hardware Rework Guide for MIMXRT595-EVK with AW-CM358MA

This section is a brief hardware rework guidance of the Ethermind Bluetooth stack on the NXP i.MX MIMXRT595-EVK board and AW-CM358MA. The AW-CM358MA user guide is available here. The hardware rework has one part:

- Debug console serial rework

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

24.1 Hardware rework

Debug console serial rework:

- Connect J39 with external power.
- JP4 1-2
- J27 1 - TX of USB to serial converter
- J27 2 - RX of USB to serial converter

Figure 61. MIMXRT595-EVK

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

25 Hardware Rework Guide for MIMXRT1040-EVK with Direct Murata M.2 Module

This section is a brief hardware rework guidance of the Edgefast Bluetooth PAL on the NXP i.MX MIMXRT1040-EVK board and the Murata 1XK or 1ZM solution - direct M.2 connection to Embedded Artists' EAR00385 (1XK) or EAR00364 (1ZM) M.2 modules.

The hardware rework has two parts:

- HCI UART rework
- PCM interface rework

25.1 Hardware rework

1. HCI UART rework
   - Solder R93 and R96
2. PCM interface rework
   - Solder R70 and R79; remove R76 and R86; Connect J80.

Note: *Make sure to disconnect J80 when debugging. Otherwise, the debugger downloading fails.*

26 Revision history

This table summarizes revisions to this document.

Table 25. Revision history

Revision number	Date	Substantive changes
0	10 June 2021	Initial release
1	08 September 2021	Updated for MCUXpresso SDK 2.10.1
2	01 December 2021	Updated for MCUXpresso SDK 2.11.0
3	11 April 2022	Updated Legal information for MCUXpresso SDK 2.12.0
4	08 August 2022	Updated for MCUXpresso SDK 2.12.1
5	28 November 2022	Added chapters Section 16 and Section 17.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

27 Legal information

27.1 Definitions

Draft — A draft status on a document indicates that the content is still under internal review and subject to formal approval, which may result in modifications or additions. NXP Semiconductors does not give any representations or warranties as to the accuracy or completeness of information included in a draft version of a document and shall have no liability for the consequences of use of such information.

27.2 Disclaimers

Limited warranty and liability — Information in this document is believed to be accurate and reliable. However, NXP Semiconductors does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. NXP Semiconductors takes no responsibility for the content in this document if provided by an information source outside of NXP Semiconductors.

In no event shall NXP Semiconductors be liable for any indirect, incidental, punitive, special or consequential damages (including - without limitation - lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort (including negligence), warranty, breach of contract or any other legal theory.

Notwithstanding any damages that customer might incur for any reason whatsoever, NXP Semiconductors' aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the Terms and conditions of commercial sale of NXP Semiconductors.

Right to make changes — NXP Semiconductors reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions, at any time and without notice. This document supersedes and replaces all information supplied prior to the publication hereof.

Suitability for use — NXP Semiconductors products are not designed, authorized or warranted to be suitable for use in life support, life-critical or safety-critical systems or equipment, nor in applications where failure or malfunction of an NXP Semiconductors product can reasonably be expected to result in personal injury, death or severe property or environmental damage. NXP Semiconductors and its suppliers accept no liability for inclusion and/or use of NXP Semiconductors products in such equipment or applications and therefore such inclusion and/or use is at the customer's own risk.

Applications — Applications that are described herein for any of these products are for illustrative purposes only. NXP Semiconductors makes no representation or warranty that such applications will be suitable for the specified use without further testing or modification.

Customers are responsible for the design and operation of their applications and products using NXP Semiconductors products, and NXP Semiconductors accepts no liability for any assistance with applications or customer product design. It is customer's sole responsibility to determine whether the NXP Semiconductors product is suitable and fit for the customer's applications and products planned, as well as for the planned application and use of customer's third party customer(s). Customers should provide appropriate design and operating safeguards to minimize the risks associated with their applications and products.

NXP Semiconductors does not accept any liability related to any default, damage, costs or problem which is based on any weakness or default in the customer's applications or products, or the application or use by customer's third party customer(s). Customer is responsible for doing all necessary testing for the customer's applications and products using NXP Semiconductors products in order to avoid a default of the applications and the products or of the application or use by customer's third party customer(s). NXP does not accept any liability in this respect.

Terms and conditions of commercial sale — NXP Semiconductors products are sold subject to the general terms and conditions of commercial sale, as published at http://www.nxp.com/profile/terms, unless otherwise agreed in a valid written individual agreement. In case an individual agreement is concluded only the terms and conditions of the respective agreement shall apply. NXP Semiconductors hereby expressly objects to applying the customer's general terms and conditions with regard to the purchase of NXP Semiconductors products by customer.

Export control — This document as well as the item(s) described herein may be subject to export control regulations. Export might require a prior authorization from competent authorities.

Suitability for use in non-automotive qualified products — Unless this data sheet expressly states that this specific NXP Semiconductors product is automotive qualified, the product is not suitable for automotive use. It is neither qualified nor tested in accordance with automotive testing or application requirements. NXP Semiconductors accepts no liability for inclusion and/or use of non-automotive qualified products in automotive equipment or applications.

In the event that customer uses the product for design-in and use in automotive applications to automotive specifications and standards, customer (a) shall use the product without NXP Semiconductors' warranty of the product for such automotive applications, use and specifications, and (b) whenever customer uses the product for automotive applications beyond NXP Semiconductors' specifications such use shall be solely at customer's own risk, and (c) customer fully indemnifies NXP Semiconductors for any liability, damages or failed product claims resulting from customer design and use of the product for automotive applications beyond NXP Semiconductors' standard warranty and NXP Semiconductors' product specifications.

Translations — A non-English (translated) version of a document, including the legal information in that document, is for reference only. The English version shall prevail in case of any discrepancy between the translated and English versions.

Security — Customer understands that all NXP products may be subject to unidentified vulnerabilities or may support established security standards or specifications with known limitations. Customer is responsible for the design and operation of its applications and products throughout their lifecycles to reduce the effect of these vulnerabilities on customer's applications and products. Customer's responsibility also extends to other open and/or proprietary technologies supported by NXP products for use in customer's applications. NXP accepts no liability for any vulnerability. Customer should regularly check security updates from NXP and follow up appropriately.

Customer shall select products with security features that best meet rules, regulations, and standards of the intended application and make the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP.

NXP has a Product Security Incident Response Team (PSIRT) (reachable at PSIRT@nxp.com) that manages the investigation, reporting, and solution release to security vulnerabilities of NXP products.

27.3 Trademarks

Notice: All referenced brands, product names, service names, and trademarks are the property of their respective owners.

NXP — wordmark and logo are trademarks of NXP B.V.

Bluetooth — the Bluetooth wordmark and logos are registered trademarks owned by Bluetooth SIG, Inc. and any use of such marks by NXP Semiconductors is under license.

EDGEFAST — is a trademark of NXP B.V.

All information provided in this document is subject to legal disclaimers.

© 2022 NXP B.V. All rights reserved.

Contents

1 Hardware Rework Guide for MIMXRT1060-EVKB and AW-AM510MA 2
1.1 Hardware rework 2
2 Hardware Rework Guide for MIMXRT1060-EVKB and AW-CM358MA 5
2.1 Hardware rework 5
3 Hardware Rework Guide for MIMXRT1060-EVKB and AW-AM457-uSD 9
3.1 Hardware rework 9
4 Hardware Rework Guide for MIMXRT1060-EVKB and AW-CM358-uSD 14
4.1 Hardware rework 14
5 Hardware Rework Guide for MIMXRT1060-EVKB and Murata uSD-M.2 Adapter 19
5.1 Hardware rework 19
6 Hardware Rework Guide for MIMXRT1060-EVKB and AW-AM510-uSD 22
6.1 Hardware rework 22
7 Hardware Rework Guide for MIMXRT1060-EVK and AW-AM457-uSD 24
7.1 Hardware rework 25
8 Hardware Rework Guide for MIMXRT1060-EVK and AW-CM358-uSD 27
8.1 Hardware rework 27
9 Hardware Rework Guide for MIMXRT1060-EVK and Murata uSD-M.2 Adapter 31
9.1 Hardware rework 31
10 Hardware Rework Guide for MIMXRT1060-EVK and AW-AM510-uSD 32
10.1 Hardware rework 32
11 Hardware Rework Guide for MIMXRT1170-EVK and AW-AM457-uSD 34
11.1 Hardware rework 34
12 Hardware Rework Guide for MIMXRT1170-EVK and AW-CM358MA 36
12.1 Hardware rework 36
13 Hardware Rework Guide for MIMXRT1170-EVK and AW-AM510MA 38
13.1 Hardware rework 38
14 Hardware Rework Guide for MIMXRT1170-EVK with Direct Murata M.2 Module 40
14.1 Hardware rework 40
15 Hardware Rework Guide for MIMXRT1170-EVK and Murata uSD-M.2 Adapter 41
15.1 Hardware rework 41
16 Hardware Rework Guide for MIMXRT1170-EVKB and Murata 1XK M.2 Adapter 43
16.1 Hardware rework 43
17 Hardware Rework Guide for MIMXRT1170-EVKB and Murata 1ZM M.2 Adapter 46
17.1 Hardware rework 46
18 Hardware Rework Guide for MIMXRT685-EVK and AW-AM457-uSD 50
18.1 Hardware rework 50
19 Hardware Rework Guide for MIMXRT685-EVK and AW-CM358-uSD 53
19.1 Hardware rework 53
20 Hardware Rework Guide for MIMXRT685-EVK and AW-AM510-uSD 55
20.1 Hardware rework 55
21 Hardware Rework Guide for MIMXRT685-EVK and Murata uSD-M.2 Adapter 57
21.1 Hardware rework 57
22 Hardware Rework Guide for MIMXRT595-EVK with Direct Murata M.2 Module 59
22.1 Hardware rework 59
23 Hardware Rework Guide for MIMXRT595-EVK and AW-AM510MA 60
23.1 Hardware rework 61
24 Hardware Rework Guide for MIMXRT595-EVK with AW-CM358MA 62
24.1 Hardware rework 63
25 Hardware Rework Guide for MIMXRT1040-EVK with Direct Murata M.2 Module 64
25.1 Hardware rework 64
26 Revision history 64
27 Legal information 65

Please be aware that important notices concerning this document and the product(s) described herein, have been included in section 'Legal information'.

© 2022 NXP B.V. **All rights reserved.**

For more information, please visit: http://www.nxp.com

Date of release: 28 November 2022
Document identifier: HWRGEFBTPALUG